Next Gen Gameplay

Manual for version 0.18.0 (beta), October 2026

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Part 1, for dummies, says what the mod does and how to use it, first the basics and then feature by feature. Part 2, the technical reference, gives the models behind it: the formulas and constants and where each number comes from, how the mod gets into the game and reads and writes its memory, and what it cannot do.

Part 1. For dummies

Getting started

1. What it is

A gauge cluster for the truck you drive, drawn over the game in the game's own font: speedometer, tachometer with the gear, fuel with the range, coolant and oil temperature, boost, brake air and up to ten lamps. The game has no engine speed, no range, no temperatures, no boost and no brake air, so the mod works those out from what the truck does, and the temperatures can cost the engine, as real ones do. Two switches go further: an automatic gearbox of the mod's own, and an engine model that changes how the truck pulls.

2. What you need

- SnowRunner for Windows: Steam, Epic Games Store or Microsoft Store. It is made on the Steam build; on other builds it finds its places by the game's code, looks again for a minute where it cannot, and stands down

where it still cannot.

- An ASI loader, which loads the mod: the `version.dll` in the release, the same file as Next Gen Tire Inflation's.
- ReShade 6.8.0, which comes in the zip as `dxgi.dll` and `ReShade.ini`, the same files as Next Gen Tire Inflation's, for compatibility with overlays such as RivaTuner, Discord, Nvidia and Steam. It shows nothing of its own; Home opens its window. A ReShade already in the game works as it is (any ReShade from 6.8.0 on with add-on support). Without ReShade the game can crash with the RivaTuner or Discord overlay or with multiple monitors, and the mod says so at the start.

3. Install

1. Close the game.
2. Copy `Gauges.asi`, `version.dll`, `dxgi.dll` and `ReShade.ini` into the game's `Sources\Bin` folder, next to `SnowRunner.exe`. A `version.dll` already there from Next Gen Tire Inflation is the same loader: keep it, and its `dxgi.dll` and `ReShade.ini` are the same files as this release's.
3. Start the game. F2 opens the NEXT GEN menu, which has a Gameplay tab.

> **Already using ReShade?** Choose Skip for `dxgi.dll` and `ReShade.ini` when Windows asks, and yours stays. If the mod then says the ReShade in the game takes no add-ons, copy in `dxgi.dll` alone. To update, copy the new file over the old one. Up to 0.15.11 the mod was `Gauges.addon64`, loaded by ReShade: delete it when you install `Gauges.asi`; with two files of the mod only one runs, and the log says which. `Gauges.ini` keeps working.

4. What you see

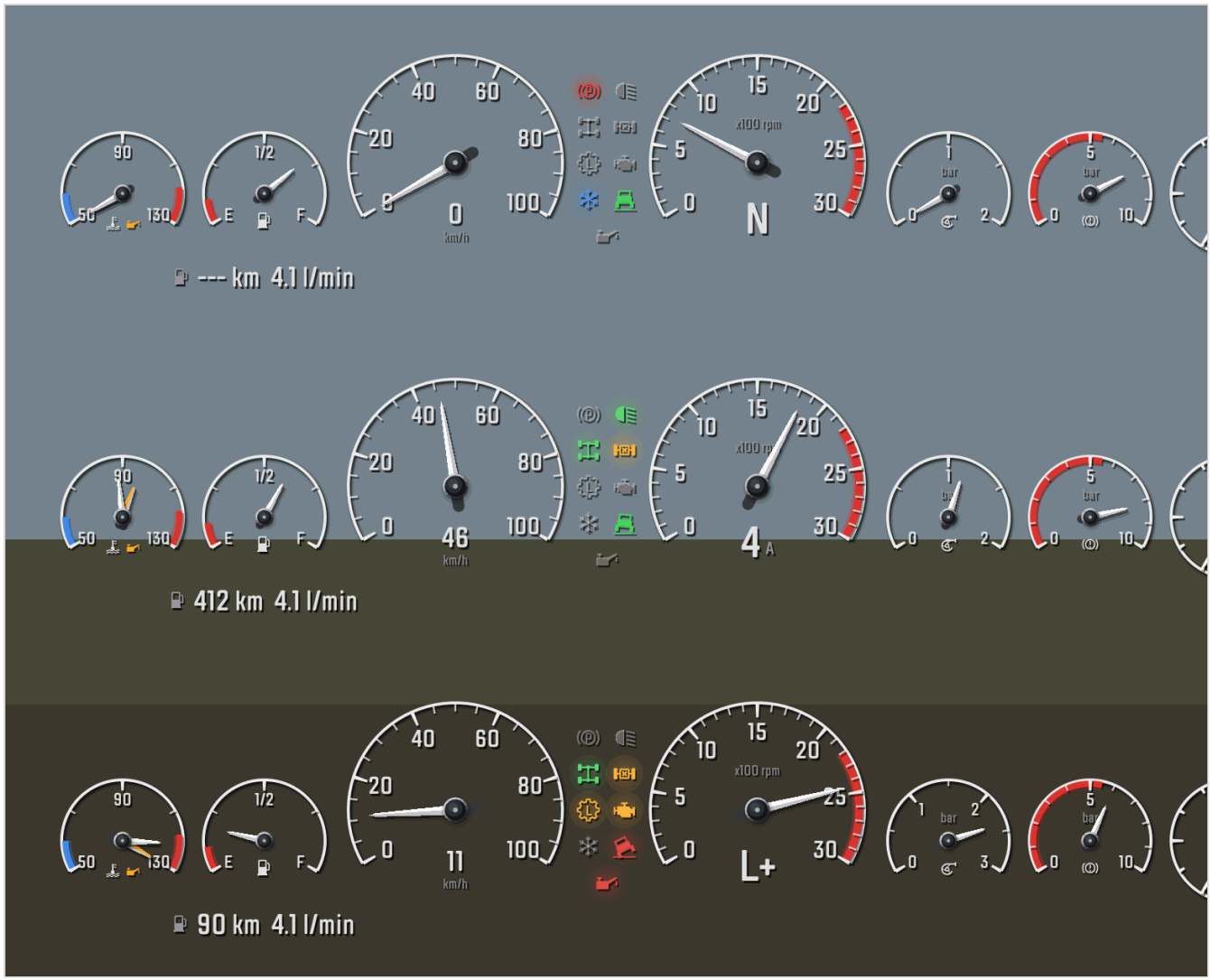


Figure 1. The default layout and style, in three states: parked with the engine cold, cruising in automatic, and a hot, tilted truck in L+.

From the left: coolant (with the oil as an amber needle), fuel with the range under it, speed, the lamps, the engine speed with the gear, boost, brake air when it is switched on, and a clock with the game's time of day and the outside temperature in its face. The cluster shows only while you drive: not on loading screens, in the pause menu, on the map or in the garage.

5. The settings worth knowing

All in the Gameplay tab of the NEXT GEN menu; they save themselves into `Gauges.ini`. F2 opens and closes the menu; to give it another key, click the key's name at the menu's top right and press the new one. The key is kept in `NextGen.ini`, which the Next Gen mods share, and while the menu is open the game gets no mouse and no keys, and the cluster's toggle key and pad toggle rest. With Next Gen Graphics installed the menu also has a Graphics tab. It switches Graphics' effects while you play. Its F keys are off then; KeysWithMenu=1 in SnowRunnerShadows.ini keeps them.

- **Layout, style, size and opacity**: a strip along the bottom or a block in the corner; solid, tinted or open dials.
- **Units**: km/h or mph, bar or psi, Celsius or Fahrenheit.
- **Truck class**: truck, scout or heavy, which sets the dials' scales. Set it while driving a truck and it is kept for that kind of truck. A scale number set by hand stays when the class changes; Class beside it gives it back.
- **Damage**: on by default. Overheating, or working a cold engine hard, adds to the game's own engine damage. Switch it off to only watch.

- **The gearbox** (`Gearbox`, 0 by default): 1 is the mod's automatic in A, smarter shifting with a torque converter and a kickdown; 2 is the mod's own gearbox on the keyboard, a pad, a wheel or a shifter, with a clutch if you want one (section 14). Tell it what you drive with: wheel pedals, a pad or the keyboard.
- **Engine to the game** (`Powertrain=1`, off by default): the truck pulls with the mod's engine. `EngineShare` sets how much; lower it if a truck pulls too hard.
- **Creep at idle** (`Creep=1`, on by default): an idling truck in gear moves, an automatic to walking pace, a manual at about the gear's idle speed; hold the brake to stand. Stuck in a gear picked by hand the engine stalls.
- **Outside by map**: the temperature follows the map and the hour of the game's day.

6. When something is wrong

- `Gauges.log` in `Sources\Bin` says what the mod found and did. Every earlier start's log is kept in `Gauges\logs`, so none is lost.
- When another program holds the game's swap chain in a way the mod cannot draw on, `Gauges.log` names that program.
- The game crashes without ReShade with the RivaTuner or Discord overlay or with multiple monitors: put ReShade in from the release (Install, step 2). With ReShade the same PCs ran fine.
- Support, discussion and ideas: the author's channel on EnthusiastRunner's Discord, discord.com/invite/MssDb2Pvcj, or an issue on GitHub for a bug report; `GrabDumps.bat` from the zip gathers the logs, the ini files and the crash dump into one zip for it.
- To remove the mod, delete `Gauges.asi`, `Gauges.ini`, `Gauges.log` and the `Gauges` folder from `Sources\Bin`. `NextGen.ini` and `version.dll` go too when no other Next Gen mod is left. ReShade from the release goes the same way when nothing else uses it: `dxgi.dll`, `ReShade.ini` and `ReShade.log`.
- Leave `Damage` off in co-op; it has not been tried there.

7. The dials one by one

- **Speed**, forwards and backwards.
- **Gear**: its number with an A in automatic, H for the high gear, L, L+ and L- for the low range, R and N.
- **Engine speed.** Worked out from the driven wheels, the gear and the pedal. The needle climbs through each gear to the red zone and falls at the shift, rises under the pedal when the truck is stuck, revs freely in neutral, and spinning wheels take it into the red zone. It rises from 0 while the engine starts and runs down when it stalls or stops.
- **Fuel** as a share of the tank, and the **range**: how far the fuel goes at the litres per kilometre burned over the last three kilometres. It shows `---` until the truck has driven half a kilometre; fuel burned standing still comes off the range without spoiling the rate. SnowRunner's trucks are thirsty: a scout burning 10 litres a kilometre off road gets about 7 km from a 75 litre tank. After the range the **fuel use**: the litres a minute the game burns now, smoothed over a second (gallons an hour with Mph; `ShowFuelUse`, the tab's Fuel use).
- **Coolant and oil** (section 9), **boost** (section 10), and **brake air**: the engine's compressor fills the tanks to 8.2 bar and starts again at 6.9; each press of the brake and letting the handbrake go draw from them. Scouts brake by hydraulics: the air dial is hidden on scout scales.

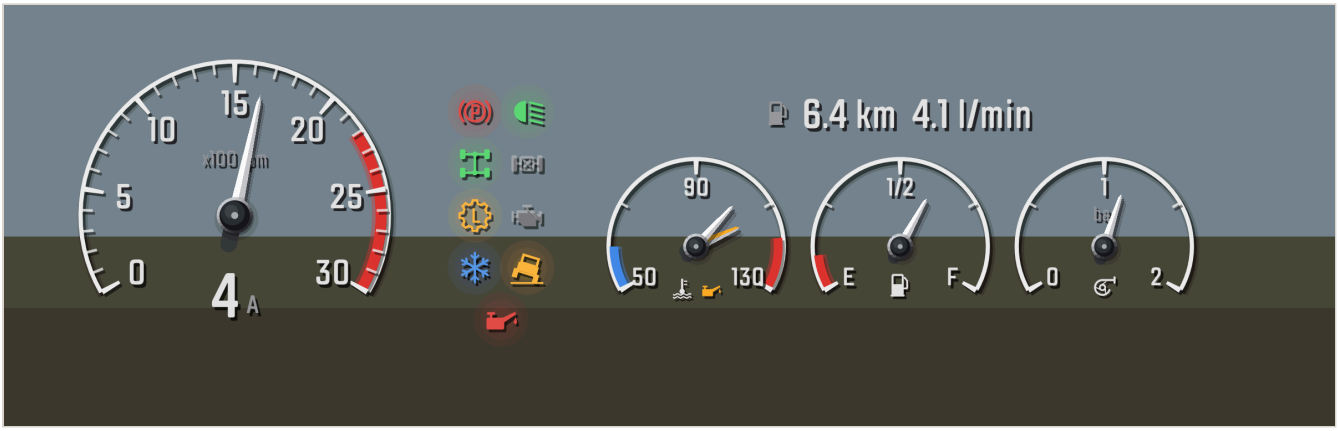


Figure 2. A close look at the open style.

8. The lamps

Table 1. Lamps

Lamp	Lights when
Handbrake, headlights, all wheel drive, differential lock, low range	The truck's own state
Engine damage	The game's engine damage is above half of what the engine takes; blinks while the mod adds damage and for two seconds after
Cold engine	Coolant below 60 °C
Oil	Blinks from 120 °C oil, with a warning every four seconds
Tilt	A small truck that leans as yours does: green level, amber from 15° to a side, red from 25°, or from 35° up or down a slope; a bump does not light it
Tire (with the tire mod)	The pressure mode's letter; amber at Low and Reduced, green at Increased, blinking while the pressure moves, red while tires take damage or one is worn out

9. Coolant, oil and engine damage

Coolant

Worked out from the fuel the engine burns and how fast the truck drives. A real engine's coolant takes about a quarter to a third of the fuel's heat; the fuel goes with the pedal and the engine speed together, so floored at the red zone makes the most heat and coasting with the foot off next to none. In neutral the engine makes a quarter of the heat.

- A cold engine starts at the outside temperature and reaches 85 °C after about three minutes of driving. The thermostat then holds about 86 °C at idle and near 100 °C working hard at speed.
- The red zone starts at 118 °C, reached only working hard at a crawl: floored at a standstill with the wheels spinning it takes a minute and a half from 90 °C. From about 7 km/h up the needle stays out of the red zone.
- Water cools it: from about half a metre at the truck's front, and with the engine in it (1.2 m and more) a hot engine is back at 100 °C within half a minute.
- A stopped engine cools from 90 °C to 40 °C in a quarter of an hour, five times as fast in deep water. Each kind of truck keeps its own temperature while the game runs. After a loading screen the truck that drives on keeps its heat (a travel, a border crossing, a reload); a game start, the garage, a recovery or another truck makes every engine cold again.

Oil

The oil takes the heat of friction in the bearings and pistons and gives it to the coolant through the oil cooler, so it follows the coolant slowly and runs a few degrees over it: about 7 over at cruise and 14 floored at the red zone. Driven as usual it is at 85 °C after about five minutes, never before the coolant. It warns from 120 °C (`OilWarnC`).

Engine damage

- In the red zone an engine loses 10% of what it can take every minute at 125 °C, nothing at 118 °C (`HotDamageFrom`, the tab's Damage from, moves that start between 100 and 124 °C): floored at a standstill with the wheels spinning, an engine is used up about five minutes after it enters the red zone.
- Below 60 °C, working hard (burning more than half the fuel it burns floored at the red zone) costs up to 5% a minute, less the warmer and the lighter the work. A cold start floored at the red zone costs 2 to 3%; driving about at moderate revs costs nothing.
- The tab's sliders set both rates: up to 100% a minute for overheating and up to 60% for a cold engine. Gauges.ini takes up to 100 for both.
- All of it is the game's own engine damage: a garage repairs it, and past the engine's threshold the game takes torque away and burns more fuel. The mod stops one point short of what the engine takes and never stops the engine itself.

Table 2. Heat settings

Setting	Default	What it changes
WarmUpMinutes	3	Minutes from 15 °C to 85 °C driven as usual
OverheatMinutes	1.5	Minutes from 90 °C into the red zone, floored and stuck
CoolDownMinutes	15	Minutes a stopped engine takes from 90 °C to 40 °C
AirFullKmh	30	Speed from which the radiator gets all the air driving gives
OilWarmUpMinutes, OilWarnC	5, 120	Oil warm-up time; too hot from
HotDamagePercentPerMinute, ColdDamagePercentPerMinute	10, 5	Damage rates; 0 switches each off
Damage	1	The temperature costs the engine

Each of the three times moves one thing only; the temperatures the engine settles at stay. The outside temperature moves those: in frost the engine warms slowly and cannot be overheated, on a hot day it overheats sooner.

10. Boost and the turbo

The mod runs a turbo of its own. Its rotor spools by the fuel the engine burns and how fast it turns: slowly low down and at a light foot, quickly high up. A truck's diesel takes about 2 s to its boost at 1100 rpm and under a second at 1760. Floored, a truck's diesel makes all its boost from about 1300 rpm and a scout's petrol from about 1650; at part load the boost goes by the fuel. On a short lift the rotor keeps turning, so the boost is back at once; it runs down over a second and a half. While the turbo spools the engine has the torque the air allows, about 0.43 of it at the first instant on a 1.8 bar truck; once the boost has built it has all of its curve. In neutral the free engine makes little boost, and on the governor it falls with the fuel.

Trucks and heavy trucks are diesels and scouts petrol, unless an engine's profile says otherwise (`Fuel`, section 15). The dial reads the manifold's pressure: a diesel's from 0, a petrol engine's from -1 bar (-15 psi), as its throttle holds the manifold in vacuum at idle (about -0.7 bar) and at a light foot. It ends at 2 bar (30 psi), or at 3 bar (45 psi) for a turbo of more than 2.

Table 3. Turbo profiles

BoostProfile	Character
0 standard	The class's own, as above
1 small and quick	Full sooner, half the lag
2 big and slow	Full later, half as much lag again
3 two stages	Full sooner with 1.3 times the boost, a short dip at the changeover

`BoostMax` sets how much boost the turbo reaches (0.3 to 3 bar). A naturally aspirated engine (section 16) has no boost, no dial and no turbo sounds.

11. Scales by truck class and per truck

Table 4. Scales by class

	Truck	Scout	Heavy
Idle	650 rpm	800 rpm	600 rpm
Red zone from	2200 rpm	5500 rpm	1900 rpm
Tachometer top	3000 rpm	7000 rpm	2500 rpm
Speedometer top	100 km/h	120 km/h	80 km/h
Boost	1.8 bar	1 bar	2 bar

Each kind of truck can have its own seven: the class, idle, the red zone, the tachometer's top, the speedometer's top, the boost and the turbo profile. Change one on the tab while you drive a truck and all seven are kept for that kind of truck, in a file of its own in `Gauges\trucks`, named as the game's files name the truck (a modded truck's `name/mod id` becomes `name@mod id.ini`). The file can be shared and dropped into another player's folder. "Back to the plain scales" gives them up. Every other truck shows the plain scales of `[Gauges]`.

```
[Truck]
Kind=chevrolet_ck1500
TachClass=1
RedlineRpm=4500
BoostProfile=2
```

A truck without its own class shifts, in the mod's automatic, by the class of its weight from the game's files: a scout up to 6 tonnes, a truck under 16, a heavy truck above. Some modded trucks carry a weight that is too high and land in the wrong class: set their class by hand.

12. The outside temperature by map

The mod reads which map you are on from the game's save (Steam, Epic and Microsoft Store saves) and takes the temperature from a section named after the map or its region, using the game's English names; a modded map goes by its id.

```
[Map:Alaska]
AmbientC=-12

[Map:Scandinavia]
DayC=3
NightC=-12
```

The air follows the game's day: coldest just before sunrise, warmest in the afternoon. A section with only `AmbientC` swings 10 °C around it. Without a section, or before the save is read, `AmbientC` of `[Gauges]` is in force; `AmbientByMap=0` uses it on every map. While the map's temperature is in force, the tab's slider sets it in the section it comes from. The tire mod reads the same temperature.

13. The mod's automatic (Gearbox=1)

The game's automatic shifts up at fixed speeds whatever the pedal, never shifts down under load until the engine bogs, and then drops to first. The mod's automatic takes the gears while the game's shifter is in A:

- A torque converter: floored at a standstill the engine stops at the stall speed (0.8 of the red zone, at most 2600 rpm) and climbs from there through the launch; on the move it follows the road speed with a little slip. A press never opens a locked converter: one that needs more pull gets a shift down. At idle in gear it creeps to about walking pace with the needle at idle (`Creep`, section 14). It locks up from first gear above its lock point and stays locked on a lift, so the engine brakes the truck; it opens near idle, at a stop and for the launch.
- Shifts up later the further the pedal is down. A light foot held steady cruises: it shifts up as soon as the next gear lands at 0.8 of the engine's peak torque speed or over, so the engine settles low. Lifting at speed shifts up a gear at a time; rolling downhill or slowing hard it keeps the gear. Coasting holds the gear down to 0.15 of the way from idle to the red zone (0.25 with the brake on, so the engine brakes), then shifts down, gears at a time when the truck slows hard. Floored it takes the gear that pulls the most in the game.
- On a climb it shifts down where floored would and holds the gear a few seconds, so it does not hunt.
- A kickdown is a push from part throttle to the floor: it drops at once to the gear that pulls the most, as far as the engine can take. Wheelspin and locked wheels do not shift it.
- `Input` fits the kickdown to what you drive with: on a pad the push has to come from under three quarters; the keyboard has no part throttle, so a held key shifts as floored and kicks nothing down. With the keyboard a lift shifts up only once the key has been let go for 2 seconds, and if the next press takes that shift back, lifts shift nothing up for 10 seconds (then 20, up to a minute), so tapping the key to hold a speed keeps the gear.
- Picking a position by hand, L+ among them, gives the gearbox back to you until A is picked, which takes it again. So the automatic never shifts out of L+, and the game keeps the diff lock there.

14. The mod's own gearbox (Gearbox=2)

With `Gearbox=2` the mod's gearbox takes the shifter from the moment you drive a truck, and the game's own shifts are refused. Its positions are A (the automatic of section 13), the numbered gears, N, R, H and the low range L-, L and L+. Up and down walk the row R, N, L-, L, L+, 1, 2 and on to the top gear, as SMT's did (L+ is the

first gear at full power); a gear key sets its gear; A hands the choice to the automatic until another position is picked. A or a gear picked from R while the truck still rolls back waits in N until it stands.

- **Bindings**: each action has two, set in the tab's Gearbox section by a click and then a press. A control already held or a hat already pointing when you click is left alone until you let it go. A binding is a key, a pad's buttons together or a stick's direction with a button (LB and the right stick up is `Pad LB+RSUp`), a wheel's or shifter's button or hat, or a pedal's axis for the clutch. The game steers with the left stick and turns the camera with the right, so bind a direction with a button held. The defaults are on the number pad: Num 8 up, Num 2 down, Num 5 N, Num 0 R and Num 7 A. A quick tap shifts. The first time the mod finds `SMT.ini` and no binding of its own, it takes SMT's keyboard bindings and options over once (its IMMERSIVE MODE as DownStaysForward).
- **Locks**: the differential lock and all-wheel drive on a key, on every truck, the upgrade fitted or not, at every `Gearbox` level. No key by default.
- **Options**, off by default: `RequireClutch` (a shift into a gear with the clutch not held: at about 5 km/h or more the gear grinds and stays out in N, standing or crawling it stalls the engine), `RequireGearHeld` (N the moment no gear key is held, for an H shifter), `SkipNeutral` (from L- to reverse and back without stopping in N), `DownStaysForward` (Gear down stops at L-, the lowest forward gear, N and R only on their own keys or the game's shifter, as SMT's immersive mode did; it wins over SkipNeutral).
- **Damage**, with `Damage` on: a stall from a shift without the clutch costs the gearbox `StallDamagePercent` of what it takes (5 by default). A clutch slipping under load heats up and, once red, costs the gearbox too (`ClutchBurn`, on by default; its pace is `ClutchHeatPercent`, 60). A box at the top left warns: amber CLUTCH HOT, then red CLUTCH DAMAGE. The clutch heats fully only at a crawl; from 15 km/h up a lugging engine stalls the game's own way instead. A repair clears the heat. A box of the same shape warns for the engine: amber ENGINE HOT, red ENGINE DAMAGE.
- **Creep** (`Creep=1`, on by default, at every `Gearbox` level): an idling truck in gear moves and stands only on the brake or the handbrake. Through a converter (the game's automatic or A, R too) it creeps to about walking pace, and nothing from 0.87 of the gear's idle speed (the wheels' speed with the engine at idle). In a gear picked by hand, the game's or a position here with the clutch up, an idle governor holds about the gear's idle speed, less as the clutch is pressed; a gear too tall to pull at idle lugs and stalls. The governor comes in over about half a second, gives less in the low range and eases off before the wheels reach the idle speed, so in a low gear on soft ground the wheels and the needle stay steady. Stuck in a gear picked by hand, the wheels under a quarter of the gear's idle speed for a second once the game's pull has built, the engine stalls as a real one does: hold the clutch, pick neutral or brake to keep it running. In A the converter takes it and the engine never stalls. The mod raises only the throttle the game hands the wheels, for the truck you drive: the pedal, the game's engine speed, fuel and sound stay at idle. Nothing in N, with the engine off or starting, in a menu or for another truck. Its strength (0.15 of the throttle at a standstill) is a first figure; `Gauges.log` has a line as each creep begins and ends.
- **Levels**: 0 is the game's own gearbox, none of its gearbox functions hooked; 1 the mod's automatic in A with the game's shifter yours, where only the reverse the game's automatic puts in on the brake is refused; 2 the mod's own gearbox. Driven with a pad or the keyboard (`Input` 1 or 2), the game's shifter picks the position too: moved to R, N, H, A, a gear or the low range, it counts as that position's key (L+ reads as the first gear). With a wheel's pedals (`Input` 0) the game's shifts stay refused and the bindings pick the position; until a binding that can reach reverse is there, 2 runs as 1. While the NEXT GEN menu is open the gearbox stands where it is.
- The manual transmission mod (SMT) is no longer needed and not compatible: remove `SMT.addon64` or `SMT.asi` and `SMT.ini`. The mod says so when it finds it installed.
- Not tested on a wheel or shifter yet; the keyboard and a pad were driven. A downshift at speed does not yet over-rev the engine, and in a manual position the engine pulls by the mod's curve at the revs it is at, so a high gear at low revs pulls less than A does.

15. The mod's engine (Powertrain=1)

The game gives all of an engine's torque with the wheels at about half the gear's top speed and less and less to nothing at the top, whatever the engine speed. With `Powertrain=1` the truck pulls with the mod's engine instead: the torque curve against the engine speed the needle shows, doubled by the converter at its stall with the automatic. Cold oil (up to 15% at 0 °C) and a turbo without boost (35% of the torque comes from it) cost torque only where the pedal asks for more than they leave. The torque holds to about 0.92 of a gear's top speed, then falls to nothing at the top, as a diesel's governor cuts the fuel; a gear's top speed stays the game's.

`EngineShare` (50) sets how much of the mod's engine the game gets: at 50 the truck pulls half way between the game's engine and the mod's, at 100 all the mod's. With the engine handed over, the automatic shifts as a truck automatic does:

- Floored, each gear to the red zone, or sooner where the next gear already pulls harder. At a light foot it shifts up about as far along as the pedal is down, never under 0.8 of the engine's peak torque speed.
- Down where the gear below comes in clear of its own point up, so it does not shift straight back.
- One gear up at a time, each held at least 2.5 seconds of moving (a scout 1.5). The tab's "Each gear going up" slider (1 to 3 s) sets it per gearbox, in `Gauges\gearboxes\<name>.ini`. If a truck accelerates too hard, raise it, and check the truck's class.
- It sets off in the lowest gear with at least a seventh of the top gear's speed, lower where the engine is weak for the truck's mass, and takes the crawling gears only when stuck.
- A shift up that the next gear cannot hold, up a grade or through mud, is not tried again for eight seconds, then each failed try doubles the wait, up to 30.

It works through the game's torque factor, the share of torque the game leaves a damaged engine: written every pass, at most six times the game's own (the game's own with the wheels at a gear's top or past it), times the engine's Torque share where one is set, and handed back when the setting goes off, the engine stops or another truck is driven. A damaged engine still pulls less.

16. Engine profiles

Each engine can have its own file, `Gauges\engines\<name>.ini`, set from the tab's Engine section with a plot of the curve in force:

Table 5. Engine profile keys

Key	What it does
PeakTorqueRpm	Engine speed of peak torque, as the maker gives it. A turbo diesel gets a broad plateau from 0.85 to 1.5 times it, three fifths of the torque at half of it and four fifths left at the red zone
Aspiration	0 turbo, 1 naturally aspirated: no boost, all of its torque at once
Fuel	diesel or petrol in place of the class's (trucks diesel, scouts petrol): a petrol engine's boost dial shows vacuum
TorqueCurve	A whole curve as x:y points, x the engine speed over the red zone's, y the torque over its most, such as 0.3:0.7,0.5:1,1:0.75
Torque	How much of its torque the truck gets with the engine handed to the game, the game's own pull included, 10 to 100 percent (100 when not set; the tab's Torque slider). For an RLM-like engine with several times a stock engine's torque: at 25 an engine with four times the torque pulls like the stock one

The profile goes with the engine, so a truck fitted with another engine in the garage pulls by that one's. The mod also reads the engine's own record from the game: torque, responsiveness, its fastest rev change and its fuel consumption.

17. The pull test

The Gearbox section has "Test the engine's pull": on a long flat road, in fourth gear or higher, floor the pedal for seven seconds. The mod holds the gear, changes the torque factor for a second at a time (halved, raised by half, at three quarters), measures the pull and puts the factor back. Lifting the foot, braking or another gear stops it. The numbers go into the log.

18. Sounds

The turbo on a lift: on a petrol engine the throttle shuts against the boost and the air beats back through the compressor wheel, slower and fainter until it is gone; the more boost, the longer and louder, never the same twice in a row. A diesel has no throttle plate and no blow-off valve, so its turbo coasts down and its lift is quiet; `DieselFlutter=1` (Flutter on a diesel) has it flutter, and your own `turbo_whistle_1.wav` to `_4` play there. Three strengths with three recordings each. Your own WAV files in `Sources\Bin\sounds` as `turbo_high_1.wav` to `_4`, and `turbo_mid_` and `turbo_low_` likewise, replace a strength's recordings; `oil_beep_1.wav` replaces the oil warning. Each time the governor stops the compressor at the cut-out the air dryer purges with a short hiss; `air_purge_1.wav` to `_4` replace it. `SoundVolume` 0 silences them, `SoundsOnly` plays them without the cluster.

19. With the tire mod

Next Gen Tire Inflation and this mod run side by side, each with its own ini and log, their tabs in the one NEXT GEN menu; the same `version.dll` loads both `.asi` files. Gauges publishes the truck, the outside temperature, the coolant, the engine speed and the boost in shared memory (`Local\SnowRunnerGauges.State.<process id>`); the tire mod takes the temperature from it. From the tire mod's version that publishes its own state, Gauges shows the tenth lamp with the pressure mode.

20. Every setting

Table 6. Gauges.ini, [Gauges]

Setting	In the ini	Default
Layout: strip along the bottom, or block in the corner	Layout 0 or 1	strip
The speedometer and the tachometer change places, in both layouts (Swap dials)	SwapDials 0 or 1	0
Style: solid, tinted, or open	Style 0, 1 or 2	open
How dark tinted faces are	FaceOpacity 0 to 0.85	0.35
Position in 1080p pixels from the layout's place	OffsetX, OffsetY	0, 0
Size, opacity	Scale, Opacity 0.1 to 1	1, 1
mph, psi, Fahrenheit	Mph, Psi, Fahrenheit	off
Temperature, boost, air dials, lamps	ShowTemp, ShowBoost, ShowAir, ShowLamps	on, on, off, on
Clock with the outside temperature (Clock)	ShowClock	on
Fuel used a minute after the range (Fuel use)	ShowFuelUse	on
Oil: none, second needle, own dial	OilGauge 0, 1 or 2	second needle
Truck class	TachClass 0 truck, 1 scout, 2 heavy	truck
Own idle, red zone, tach top, speedo top	IdleRpm, RedlineRpm, MaxRpm, SpeedMax	0: the class's
Boost reached, 0.3 to 3 bar	BoostMax	0: the class's
Turbo profile	BoostProfile 0 to 3	standard
Tilt lamp angles	RollWarnDeg, RollDangerDeg, PitchDangerDeg	15, 25, 35
Coolant times	WarmUpMinutes, OverheatMinutes, CoolDownMinutes	3, 1.5, 15
Outside temperature, -40 to 50	AmbientC	15
Outside temperature by map	AmbientByMap	1
Radiator full air from, 10 to 80 km/h	AirFullKmh	30
Oil warm-up, too hot from	OilWarmUpMinutes, OilWarnC	5, 120
Temperature costs the engine	Damage	on
The gearbox: the game's own, the mod's automatic in A, the mod's own	Gearbox 0, 1 or 2	0
A shift without the clutch stalls; neutral while no gear key is held; L- to reverse without N	RequireClutch, RequireGearHeld, SkipNeutral	off, off, off
The gearbox's bindings, two an action	[Gearbox] Bind<Action>, Bind<Action>2	Num 8, 2, 5, 0, 7 for up, down, N, R, A
Stall damage, percent of the gearbox, 0 to 25	StallDamagePercent	5
The clutch burns; its heating pace, 10 to 300	ClutchBurn, ClutchHeatPercent	on, 60
Driven with: wheel pedals, pad, keyboard	Input 0, 1 or 2	wheel pedals
The mod's engine	Powertrain	off
Share of the mod's engine, 0 to 100	EngineShare	50
An idling truck in gear moves (Creep at idle)	Creep	on

Setting	In the ini	Default
Damage rates, 0 to 100	HotDamagePercentPerMinute, ColdDamagePercentPerMinute	10, 5
Where the hot damage starts, 100 to 124 °C (Damage from); the rate is reached at 125 °C	HotDamageFrom	118
Sound volume, 0 to 100	SoundVolume	50
Sounds without the cluster	SoundsOnly	off
A diesel's lift flutters (Flutter on a diesel)	DieselFlutter	off
Key and pad buttons that hide the cluster	ToggleKey, PadToggle	none
For work on the mod	Demo, Probe	off

21. What the mod writes, and the logs

- No game file is changed. Into the game's memory it writes only: with `Damage` on, the driven truck's engine damage, and its gearbox damage on a stall from a shift without the clutch or from a burning clutch (on the game's thread, through a hook on its truck update, with the game's damage update after); with `Gearbox=1` or `2`, the game's automatic flag, the gears through the game's own shift function, the low range's power factor and the stall seconds of a shift without the clutch; with `Powertrain=1`, the torque factor every pass; with `Creep=1` nothing in the truck: the throttle the game hands the driven truck's wheels at idle is raised on its way, through a hook on the game's wheels' drive; during the pull test, the torque factor for seven seconds. With all of them off it writes nothing. The hooks in the game's code go by `Gearbox`, and the creep's by `Creep` alone, at every level: at 0 none of the gearbox's; at 1 and 2 the eight gearbox functions, calling through; at 2 with `Damage` on also the game's truck update. Each is put in the first time the level needs it and stays until the game is started again.
- `Gauges.log` is the last start's log; at each start the previous one moves to `Gauges\logs`, named by its date and time. With `Probe=1` a long drive's log runs to 20 or 30 MB, and the folder is never emptied by the mod.
- The game's own HUD stays; its "Immersive Mode" option hides it while not in use.

Part 2. Technical reference

This part is for readers who want the workings. The game keeps no engine speed, no temperatures, no boost, no brake air and no range, so most of the cluster is models: this part gives each one as the code has it, with its formulas, every constant, and where a number comes from when the code says so; then how the mod finds the truck in the game's memory, what it reads and writes there, and what it cannot do. The numbers are those of version 0.18.0; file and line references (`models.h:153`) point into its source. "The Steam build" is the game build of 22 July 2026 (time stamp `6a607c05`); an RVA is an offset from the start of `SnowRunner.exe` in memory, in that build. The charts are the models re-run with their default settings.

> **Notation.** *p* is the pedal, 0 to 1, and *b* the brake. spin is the mean turning speed of the driven wheels in rad/s. top(*g*) is gear *g*'s top wheel speed. idle, red and max are the tachometer's idle, the start of its red zone and its end. share(rpm) = clamp((rpm − idle)/(red − idle), 0, 1) and speed(rpm) = clamp(rpm/red, 0, 1). clamp(*v*, *a*, *b*) holds *v* between *a* and *b*. A pass is one loop of the mod's thread.

22. How the mod runs

`Gauges.asi` is loaded into the game by an ASI loader (the release's `version.dll`) and draws each frame from its own hook on the game's Present (Drawing, below). It acts only inside `SnowRunner.exe`, and a mutex `Local\SnowRunnerGauges-<process id>` lets one copy run. It starts one thread that loops with a 50 ms sleep, about 20 passes a second (`gauges.cpp:2060`). Each pass reads the truck, steps every model by the time since the last pass, Δt = clamp($t - t_{last}$, 0.001, 0.2) s (0.05 s after a gap of a second or more), writes what it writes, and hands the state to the drawing. The drawing runs at the frame rate and lets the two big needles glide between passes with a 0.05 s time constant.

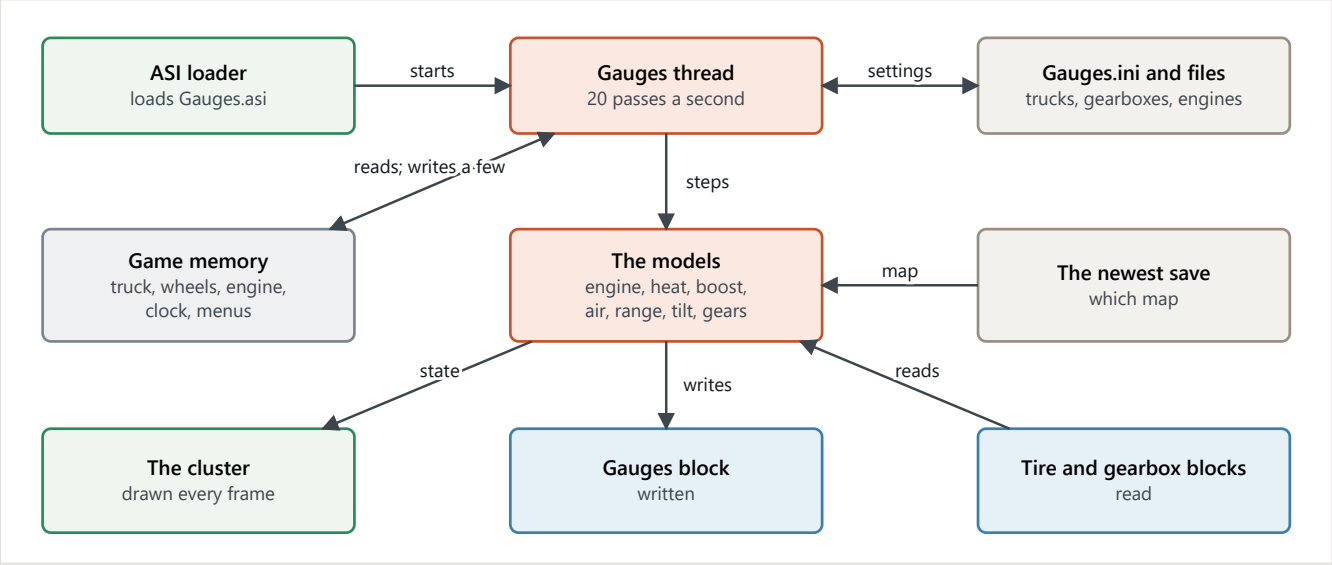


Figure 3. The parts and what flows between them.

The models only step while you drive: no menu, map, garage or viewer open, not loading, and the game's simulated clock (vehicle +0x114) having moved in the last second, so a paused game stops them. The cluster shows after half a second of driving.

23. Scales and classes

Table 7. The class scales (gauges.h:194); no stated source

Class	Idle	Red zone from	Tachometer to	Speedometer to	Boost to
Truck	650 rpm	2200 rpm	3000 rpm	100 km/h	1.8 bar
Scout	800 rpm	5500 rpm	7000 rpm	120 km/h	1 bar
Heavy	600 rpm	1900 rpm	2500 rpm	80 km/h	2 bar

A nonzero `IdleRpm`, `RedlineRpm`, `MaxRpm`, `SpeedMax` or `BoostMax`, from a truck's own file or from `[Gauges]`, replaces the class value. The scale is then made to fit (`GaugeScaleOf`, `gauges.h:198-211`): max is rounded up to a step of 1000 rpm (500 below 5000), red is held to at most max, and idle to at most 0.8 red.

The automatic and the torque curves need a class too. It is the truck file's `TachClass` if that file sets any scale, otherwise the class by the truck's mass (model +0x784, without addons, cargo or trailer): up to 6000 kg a scout, under 16000 kg a truck, from 16000 kg a heavy truck (`models.h:1282-1292`; the TUZ 108 weighs 4.1 t, the Tayga 12.0 t, a twin-steer Western Star 20.3 t). Weight cannot tell every scout from a truck: the Fleetstar (6314 kg) counts as a truck.

24. The gearbox as the game has it

The truck keeps a list of numbers, one per gear (truck action +0x58): reverse, 1 to *N*, high. The game turns a gear's number a into the gear's top wheel speed (`TruckGearTop`, `models.h:127-135`):

$$\mathrm{top(g)} = 2\mathrm{a} + 5 \text{ rad/s}, \mathrm{top(high)} = \mathrm{a} + 2 \tag{1}$$

A numbered gear's top is learned where the game's own pull reaches nothing (review 11, `GearTopStep`): a floored plateau, the pedal at 0.9 or more, +0xB4 at 0.02 or under, the power factor 1, the wheels between 0.85 and 1.15 of $2a + 5$ and within 1% of their mean for a second, gives the wheels' mean as the gear's top, kept in the gearbox's file (Top*g*) and averaged with the next plateau's; until then $2a + 5$. The high gear's $a + 2$ holds (an HX 520 sat at 15.8 to 16.3 rad/s in H against 16 in 11 floored plateaus). The testers' logs of 0.17.0 at hand had plateaus at the top of the first gear (0.92 to 1.03 of $2a + 5$) and of H (1.03) only; the review read 0.90 to 0.99 in some gears from the pull's fall.

In first gear with the automatic off, the low range scales it: top = 1 + (top − 1) · clamp(power factor, 0, 1), with the power factor at truck action +0x38. The game's low ranges are L 0.45, L− 0.2 and L+ 1; the lamp reads below 0.325 as L−, below 0.725 as L, otherwise L+.

The game does not give an engine's torque by its engine speed. It gives it by how far the wheels are toward the gear's top, read from truck action +0xB4 with the pedal floored and steady on the Defender and another truck (`models.h:1081-1088`):

$$\mathrm{GamePull(x)} = \mathrm{clamp}(\ \min(2.8\,\mathrm{x} - 0.25,\ 2.1\ (1 - \mathrm{x})),\ 0,\ 1\ ),\ \mathrm{x} = \mathrm{spin}\ /\ \mathrm{top(g)} \tag{2}$$

All of it from $x \approx 0.45$ to 0.52, falling in a line to nothing at the top. In 0.12.5's test a power factor that doubled a gear's top pulled less, so the game multiplies the torque as a gear's top is lower; at a given road speed a gear pulls $x \cdot \mathrm{GamePull}(x)$ against the others. A gear revved near its top pulls a fraction of the next one's.

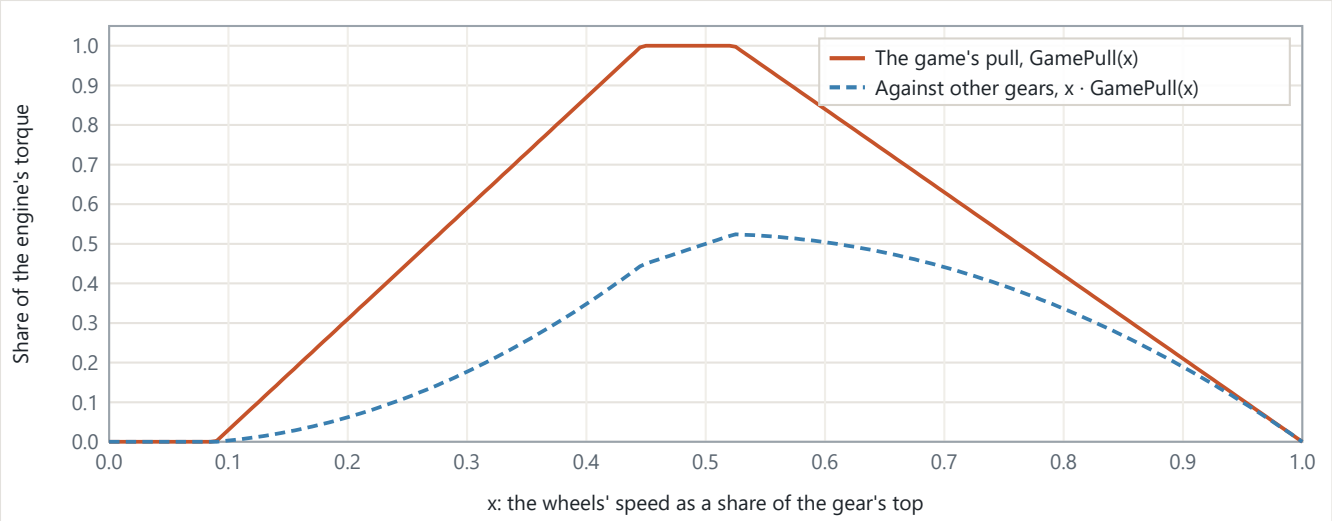


Figure 4. How the game hands out an engine's torque by the wheel speed, as the mod models it.

So the gear that pulls the most has the wheels near half its top. Floored, the automatic shifts up at the first share $x \geq 0.5$ (in steps of 0.005) where the next gear pulls 5% more, and down at the first share from 0.6 down where the gear below does (`GearUpShare`, `GearDownShare`, `models.h:1313-1328`):

$$\text{up: } (x/r) \cdot \text{GamePull}(x/r) \geq 1.05 \cdot x \cdot \text{GamePull}(x),\ r = \text{top}(g{+}1)/\text{top}(g) \tag{3}$$

Table 8. Where the automatic shifts floored with the game's own engine

Step to the next gear, top(g+1)/top(g)	Floored up at, share of red	Floored down at, share of red
1.2	0.625	0.485
1.35	0.665	0.465
1.5	0.700	0.445
1.75	0.755	0.420
2	0.805	0.390
2.5	0.870	0.340

25. Engine speed

The needle heads for a target worked out from the wheels and the pedal (`TachTarget`, `models.h:153-167`). With $q = \sqrt{p}$:

$$\text{in gear: } w = \text{clamp}(\text{spin} \,/\, \text{top}(g),\ 0,\ 1.1),\ s = 1 - \min(w,\ 1),\ t = \text{clamp}((w\ \text{red} - \text{idle}) \,/\, (\text{red} - \text{idle}),\ 0,\ 1.1),\ n = t + (w - t)\ q + 0.65\ q\ s^{1.5} \tag{4}$$

$$\text{in neutral: } n = 0.9\ q \tag{5}$$

$$\text{target} = (\text{idle} + (\text{red} - \text{idle}) \cdot n) \cdot \text{alive}^2,\ \text{alive} = \text{clamp}(1 - 0.714 \cdot \text{stall},\ 0,\ 1) \tag{6}$$

stall is the game's stall timer (vehicle +0x11C); the engine dies at 1.4 s, where alive reaches 0. The wheels at the gear's top put the needle at the red zone; floored at a standstill n = 0.65, floored in neutral 0.9. At zero pedal the wheels turn the engine in proportion, never under idle (t), so a truck that creeps at idle shows idle; the map from idle comes in with q. Starting, the target is idle × clamp(1 − timer/1.8, 0, 1) on the game's start timer (vehicle +0x730). With the engine off it is 0.

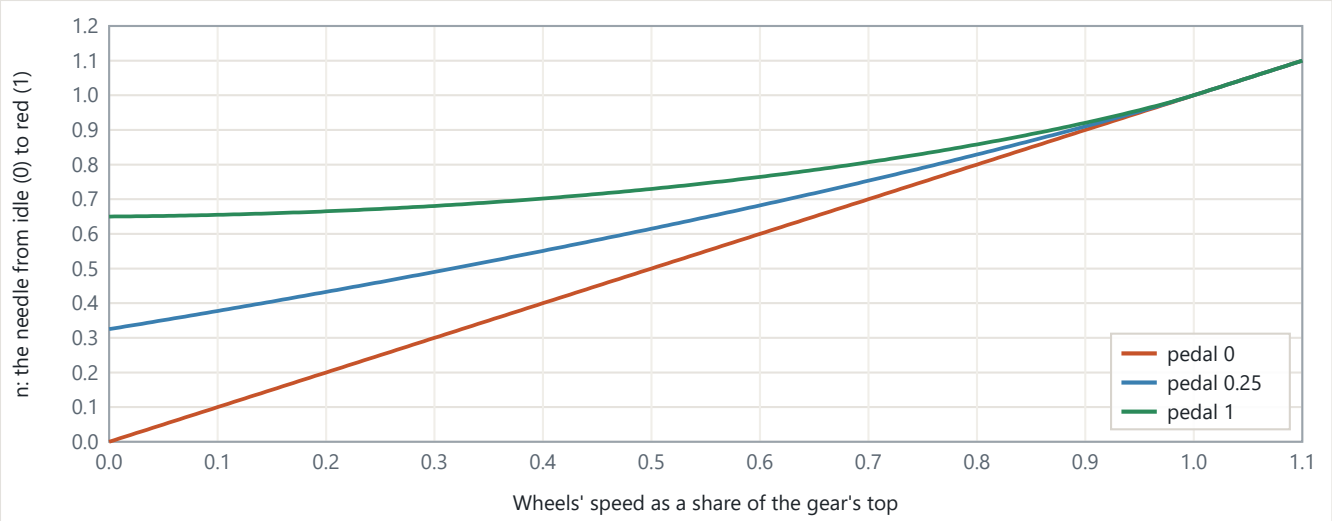


Figure 5. Where the needle heads in gear. Floored at a standstill it stands at 0.65 of the way to the red zone; wheelspin past the top takes it to 1.1.

The needle follows with a lag (`TachStep`, `models.h:178-197`):

$$\text{rpm} \leftarrow \text{rpm} + (\text{target} - \text{rpm}) \cdot (1 - e^{-\Delta t/\tau}),\ \tau = 0.18 \text{ s rising, } 0.25 \text{ s falling, } 0.4 \text{ s engine off} \tag{7}$$

While the gear changes, and for 0.3 s after, the target is × 0.85, the dip of a shift. The first pass on a truck puts the needle at the target at once. Under the mod's automatic the needle follows the torque converter's engine speed instead (see the automatic), with the same lag and no dip.

The probe also logs the game's own engine value, which the game burns fuel by (`models.h:236-260`): 0.19 at idle, 0.69 floored, scaled by the engine's damage. The cluster does not use it.

26. Torque curves and engine profiles

The mod's engine has a torque curve against its engine speed, as points *x*:*y* with *x* the engine speed over the red zone's and *y* the torque over its most, straight between points and flat below the first. Past the last point the torque falls in a line to nothing at the tachometer's end (`TorqueAt`, `models.h:923-931`).

Table 9. The class curves (models.h:887-890): trucks and heavy trucks a diesel's, broad and early; scouts a later peak

Class	Points
Truck	0.25:0.55, 0.45:0.85, 0.6:1, 0.8:1, 1:0.85
Scout	0.15:0.5, 0.35:0.75, 0.6:0.95, 0.75:1, 1:0.85
Heavy	0.25:0.6, 0.4:0.9, 0.55:1, 0.75:1, 1:0.8

An engine profile's `PeakTorqueRpm` builds the curve around the maker's peak torque speed (`EngineCurveFromPeak`, `models.h:938-961`). With P = clamp(peak/red, 0.15, 0.95):

$$\text{turbo: } (0.5P, 0.6), (0.85P, 1), (\min(1.5P, 0.95), 1), (1, 0.8) \tag{8}$$

$$\text{aspirated: } (0.4P, 0.7), (P, 1), (P + 0.5(1 - P), 0.92), (1, 0.8) \tag{9}$$

A point closer than 0.005 to the one before is left out. `TorqueCurve` gives 2 to 8 points of its own, *x* above 0 and rising, *y* from 0 to 1, or is ignored. A hand-written curve wins over the peak, the peak over the class curve.

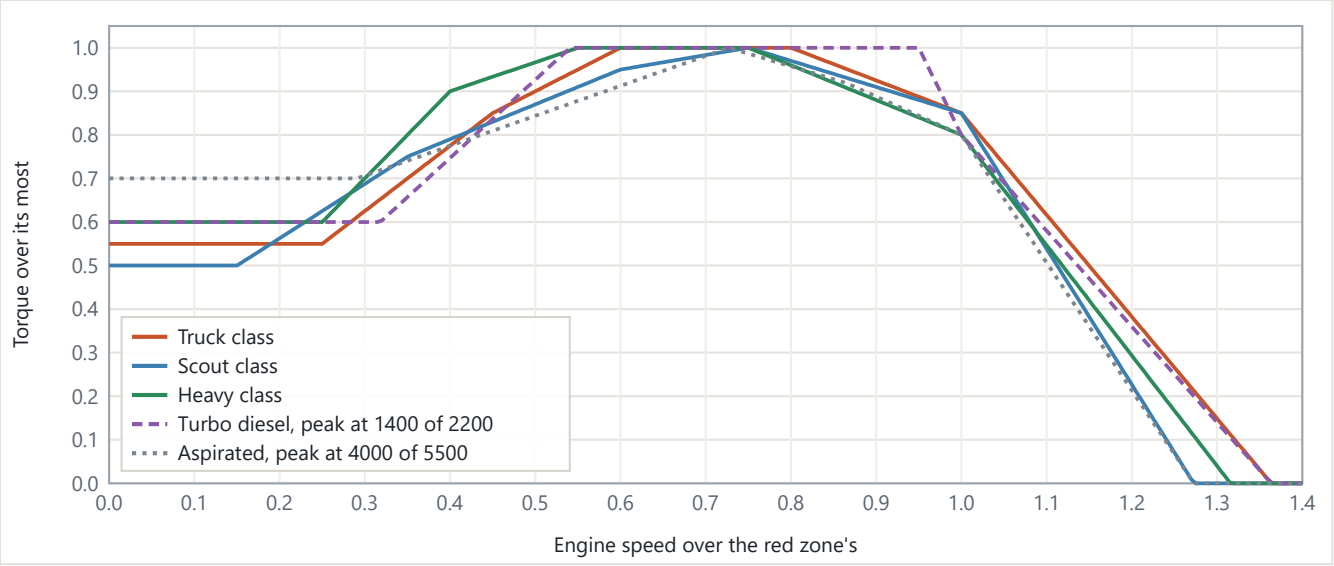


Figure 6. The torque curves: the three class curves, and two built from a peak torque speed. Past the red zone each falls to nothing at the tachometer's end.

27. The mod's engine handed to the game (Powertrain=1)

The mod's engine works out what it would give each pass (`PowertrainStep`, `models.h:1009-1022`):

$$\text{pedal}' \to \text{p}^{1.8}, \text{ from 0 to 1 in 0.6 s at the most, from 1 to 0 in 0.15 s} \tag{10}$$

$$\text{cold} = 1 - 0.15 \cdot \text{clamp}((60 - \text{T}_{\text{oil}})/60, 0, 1), \text{boost} = \text{limit (equation 34)}, 1 \text{ aspirated} \tag{11}$$

$$\text{torque} = \text{curve}(\text{speed}(\text{rpm})) \cdot \text{cold} \cdot \text{boost}, \text{power} = \text{torque} \cdot \text{pedal}' \tag{12}$$

Cold oil costs up to 15% at 0 °C and nothing from 60 °C. An aspirated engine has boost = 1. Half way down the pedal gives 0.29.

The game has no place for an engine curve, but it has a torque factor: the share of its torque a damaged engine keeps (truck action +0xCC). The mod writes that factor so the game's pull matches the mod's engine (`models.h:1160-1238`). With *x* the wheels' share of the gear's top, from the learned grip so that spinning wheels keep the game's fall:

$$\text{lift} = \text{clamp}(\text{ curve}(\text{x}) \cdot \text{ratio} \cdot \min(1, \text{cold} \cdot \text{boost} / \text{p}) / \text{GameFall}(\text{x}), 0.05, \text{R}) \text{ for } \text{x} < 1, 1 \text{ for } \text{x} \geq 1,$$
$$\text{GameFall}(\text{x}) = \text{clamp}(2.1(1 - \text{x}), 0, 1), \text{R} = 6 \tag{13}$$

$$\text{factor} = \text{game} \cdot (1 + \text{clamp}(\text{EngineShare}/100, 0, 1) \cdot (\text{lift} - 1)) \tag{14}$$

ratio is the torque converter's multiplication while the mod's automatic holds the gears, otherwise 1. game is the game's own share, at most 1; it is taken at the first write and again whenever the value read differs from the last one written (the game wrote its own, as on new engine damage), so a damaged engine still pulls less. The factor is written only when it moves by 0.005 or more, never below 0.05 and never above the roof: 6 times the game's own for a truck or a heavy truck, 3 for a scout (the class the automatic goes by), flat up to 0.7 of a gear's top and from there falling in a straight line to 1 at the top (6: 3.5 at 0.85, 1.75 at 0.955; 3: 2 at 0.85). With the fall at $2.1(1 - x)$ the engine's torque is held to about 0.86 of a gear's top (a scout's 0.80), then the push dies down to the game's own at the top, as a diesel's governor cuts the fuel. With a flat roof of 6 it held to 0.92 (at 3 times 0.84), and a vanilla Defender 90 floored on a road went from 27 to 58 km/h in 5 seconds through three gears (27 October). The fade began at 0.8 until 28 October (held to 0.88): over the last fifth the push fell so steeply that a heavy truck held near its top gear's limit at part throttle overshot into the fade, got the push back in one jump and surged, every two seconds (a tester's HX 520 at 36 to 39 km/h, the factor swinging 1.0 to 2.4). Within a gear a factor above the last one written climbs toward it by at most 1 a second (and at least 0.005 a pass, so it is written); a lower one, a new gear (the game's engaged gear) and the first write after the hand takes the truck are written at once. Near the top a small error in the fitted fall times a big factor is a big error in the torque, and a roof of 23.8 kicked a d110 at the top of 2nd to 5th gear (2026-10-25). With the wheels at the gear's top or past it (rolling downhill) the factor is the game's own: the division would be by nothing. Below about half a gear's top GameFall is 1 and nothing is divided out. The game's own share goes back once when the setting goes off, the engine stops, the pull test is armed, or another truck is driven (and to the old truck only if its action is still its own).

The pull test

On a long flat road in fourth gear or higher (the game's automatic off), floored, no brakes, at 1 m/s or more and the wheels under 0.55 of the gear's top, the test holds the gear and runs seven phases of a second each, with the factor × 1, 0.5, 1, 1.5, 1, 0.75, 1 (`models.h:1804-1944`). In each phase it fits a straight line to the speed over time by least squares (the first 0.3 s apart), slope = $(n\Sigma ty - \Sigma t\Sigma y)/(n\Sigma t^2 - (\Sigma t)^2)$, and reports each changed phase against the mean of its neighbours:

$$\text{ratio}_i = \text{slope}_i \,/\, ½(\text{slope}_{i-1} + \text{slope}_{i+1}),\ i = 1, 3, 5 \tag{15}$$

A ratio near the phase's factor means the game's pull goes with the factor. Lifting, braking, another gear or the wheels past 0.95 of the top stop it, and the factor goes back. The power factor (action +0x38) was tested the same way in 0.12.4: a write counts at once and is kept, but it moves the gear's top speed and gives no torque.

28. The mod's automatic (Gearbox=1)

The automatic is a model that runs every pass (`GearboxStep`, `models.h:1419-1652`); with `Gearbox=1` it also drives the game's gears. Its "engine mode" is on while the Powertrain factor is being written.

What it goes by

- **The pedal.** With `Input` = keyboard any pedal from 0.05 is taken as floored. A drop of more than 0.05 for up to 0.3 s is ignored: the game cuts the throttle for a moment at each shift. The pedal the shift points follow, feel, falls at most 0.5 a second: feel = max(p, feel − 0.5Δt).
- **The grip.** Learned only when it can be trusted: pedal over 0.2, no brake or handbrake, over 2.5 m/s on the ground, the wheels over 1 rad/s, steady within 5%, the wheels within 15% of each other, and no hard acceleration. It is the wheels' turning over the ground speed; it follows a lower value with a 2 s time constant and a higher one with 20 s.
- **The speeds.** spin smoothed over 0.25 s; by = grip × ground speed once the grip is known; its trend smoothed over 1 s. Foot on, the shifts go by pace = min(spin, 1.2 × ground + 1 rad/s): in drives logged in the game the wheels turned within 1.13 times the learned grip nine times in ten, so wheelspin does not shift up. A shift up with the foot on goes by min(wheels, 1.05 × by), the wheels unsmoothed. With the engine handed to the game the gears it sets off in (the first, and the start gear) differ: their shift up, floored point and limiter alike, goes by the wheels as they turn, spinning or not, while the truck goes at least 0.3 of the start gear's top over the ground. The game's pull falls by the wheel speed, so a spinning first gear pulls almost nothing and gains nothing from staying (a tester's log of 27 October: a ZIL-157 on dirt sat in first 10 to 40 s). The gear's hold still gives a launch its seconds there, and a truck stuck where it stands keeps the ground's rule. Foot off they go by the faster of the wheels and the ground.
- **Climbing.** Half a second with the pedal at 0.4 or more and the speed falling faster than 0.35 rad/s² starts 4 s in which the shift points are the floored ones.

Shift points

With point(s) = idle + (red − idle) s and input(g, v) = v/top(g) · red, the engine speed a gear would turn at:

$$\text{up}(g) = \text{light} + (\text{floored} - \text{light}) \cdot \text{feel},\ \text{light} = \text{point}(0.35),\ \text{floored} = \max(\text{up share} \cdot \text{red}, \text{light}) \tag{16}$$

$$\text{down}(g) = \text{point}(0.12) \text{ at a light foot},\ \max(\text{down share} \cdot \text{red}, \text{point}(0.12)) \text{ floored} \tag{17}$$

The shares are those of equation (3). In engine mode the points follow the mod's curve instead: the floored point is where the next gear pulls harder by the mod's curve, EngineShare and the roof (with a truck's fading roof a ZIL-131's at 2816 to 2864 rpm of a 3200 red zone; with the next gear 43/27 taller, at 0.905 of the gear's top floored with all of the engine, 0.875 at EngineShare 50), or the limiter, point(0.98), where none does (until 0.15.10 never past point(0.95), a rule from before the fade, which sat just over it); the lightest is the light-foot shift-up point, 0.8 × where the curve first reaches 0.9 of its most and no later than 0.45 red (a truck's class curve 880 rpm, a scout's 2370; from the curve's 0.98 point a scout's was 3300), and never under idle, as a car's nine-speed automatic shifts; down is the gear below's up point moved over by the step, / 1.15, and never under 1.6 idle at a light foot, falling to 1.15 idle floored (a truck's part-throttle points land at

0.55 to 0.75 of its governed speed and a modern diesel pulls cleanly to about 1000 to 1100 rpm; review 11, from 1.15 at any pedal).

Two more ways up, one gear at a time, in both modes. Cruising: a light foot (0.5 or less) held steady, the speed changing by no more than 2% of itself a second, for 2 s, goes up when the next gear would turn at max(the light-foot shift-up point, 1.1 × its point down) or over, short of the light point (a car's nine-speed automatic, filmed on the road on 2026-10-19, cruised at 1200 to 1300 rpm against a peak at about 1600). A lift at speed: the foot off and no brake, the truck neither gaining more than 2% a second (downhill: the engine brakes it) nor slowing more than 4% (mud, a climb), goes up when the next gear would turn at max(the light-foot shift-up point, 1.1 × point(0.15)) or over. With keys (`Input` 2) the key has to have been let go for 2 s first, and a lift's shift up taken back down within 6 s (the next press, the brake) keeps lifts from shifting up for 10 s, twice as long each time again, up to 60 s: a tester's keyboard drive on 0.15.8 had 19 of 23 lift upshifts taken straight back, 376 of its 412 lifts under a second. A wheel's pedals and a pad shift up at a lift as before.

Deciding

1. Standing (the first gear would turn under 0.1 red): the start gear at once.
2. A kickdown (below).
3. Past the red zone, or past what the engine can take: up, at once, through every gear that would be past it too.
4. Up: the shift has been called for 0.4 s at a light foot, 0.15 s floored (a short first gear is through in a second), and the last shift is 0.5 s past if it was also up.
5. Foot off: down when the gear would turn under point(0.15), under point(0.25) with the brake on (the same car coasted in its gear down to 1000 to 1300 rpm of a 4500 rpm red zone, 0.07 to 0.15), and the gear below under point(0.75), called for 0.3 s.
6. Foot on: down when called for 0.5 s.

Every shift pauses decisions for a second; the converter opens for the shift's own pass and may lock again on the next. In engine mode three more rules hold: a gear is kept 2.5 s of moving after a shift up (1.5 s for a scout, or the gearbox file's `GearHold`; an empty Fleetstar went 1 to 5 in two seconds), but on the limiter only 0.9 s (review 11: no truck automatic holds a gear at the governed speed while the truck accelerates); the start gear is the lowest with a top of at least a seventh of the top gear's, lower while the engine's torque per kilo is under 1.2 × the gear's top; and a shift up taken back within 6 s with the foot on is not tried again for 8 s, doubling up to 30 s (a tester's Tayga swapped first and second 18 times in six minutes on 0.15.0).

Kickdown

A push from part throttle to the floor: the pedal must have stood between 0.2 and 0.9 (0.75 for a pad, whose trigger held near its end wavers) for 0.3 s, then reach 0.9. A press from the foot off, so every key press, does not kick down. It then steps down at once, ignoring the pause, while the gear below fits and pulls 5% more by equation (2) (engine mode: while the gear below would turn under min(point(0.85), 0.92 × its up point)).

The torque converter

With input = spin/top(*g*) · red the speed the gearbox turns the engine at (`models.h:1637-1651`):

$$\text{stall speed} = \max(\ \min(0.8\ \text{red}, 2600), \text{idle}\ ), \text{stall} = \text{idle} + (\text{stall speed} - \text{idle}) \cdot \text{p} \tag{18}$$

$$\text{rpm} = \max(\text{input}, \text{idle})\ \text{locked};\ \max(1.04\ \text{input}, \text{stall} + 0.15\ \min(\text{p}/0.05, 1)\ \text{input}, \text{idle})\ \text{open} \tag{19}$$

$$\text{s} = \text{input}\ /\ \text{rpm}, \text{ratio} = 1\ \text{if s} \geq 0.87, \text{else}\ 2 - \text{s}/0.87 \tag{20}$$

The stall speeds are 1760 rpm for trucks, 2600 for scouts and 1520 for heavy trucks (Allison's converters stall at 1500 to 2000 on a heavy truck): the truck's diesel curve gives all its torque there, the heavy truck's 0.96, a scout's 5500 rpm engine 0.85. Open, the engine climbs from the stall with the input, as the converter's K-factor rises with the speed ratio, so the needle climbs through a launch; the rise comes in over the first 5% of the pedal, so at idle the engine stays at idle while the truck creeps. The ratio is 2 at a stall and falls in a line to 1 at the coupling point, 0.87. Lock points: 0.5 red with the game's own engine; in engine mode 1.65 idle at a light foot up to 0.5 red floored. In the gears it sets off in it locks only once the speed ratio is 0.92 too, so the lock drops the needle at most 8%; a floored launch stays in the converter through

first, as Allison's 1C. A shift made locked opens it for its own pass and locks it again on the next over the lock point, so from there the gears go lock to lock. It lets go under 0.4 red with the game's own engine; in engine mode under 1.4 idle with the foot on and 1.04 on a coast, never by the pedal, so a press from a low cruise keeps it locked and a press that needs more pull gets a shift down by the schedule or the kickdown (review 11: the unlock point had risen with the pedal, and a press from a low cruise flared the engine to the stall). A lift keeps it locked, as a truck's automatic does: the engine follows the wheels down and the game's own engine braking slows the truck in the gear held. Before d0940b4 it opened on every lift for a second and at every shift for the shifts' pause, never locked in first; it stalled at 0.75 red (2400 at most), then at 0.6 (2000 at most, the tight one of d0940b4), coupling at 0.85 and then 0.75: the truck felt like a rubber band (two players' reports, 28 October).

Creep at idle (Creep=1)

Once a physics step the game makes a drive throttle and a brake from the pedal field (+0x44; d740a0 in the 6ab52f5d image) and hands the throttle to the wheels' drive, `c41da0(action, throttle, dt)`, called only at c41d40. The pull +0xB4 moves toward throttle · (1 − D4 + D4 · curve(the wheels' speed)) and each driven wheel gets min(B4, throttle) · CC · D4 · the gear's torque · its share (D4 is action +0xD4, CC the torque factor), so at zero pedal every wheel gets nothing. The game's engine speed value (d73200, TruckGameRevs), its fuel and as far as is known its sound read the pedal field, not this throttle. The mod finds the call by its bytes (the action in rcx, the throttle in xmm1, the seconds in xmm2; once in the code), checks that its target is a function's start with the drive's first bytes and not another mod's jump, and detours the drive to a stub (`creep_stub.asm`) that sets xmm1 to the larger of the throttle and the mod's share for the driven truck's action only; every other call passes as it came. The share is worked out each pass (`CreepShare`) and handed over with its action as one slot:

converter: $r = \text{input} / \max(1.04\ \text{input}, \text{idle})$, $\text{share} = 0.15 \cdot (1 - \text{clamp}(r/0.87, 0, 1))$ (20a)

by hand: $\text{most} = 0.15 \cdot \min(\text{top}(g) / \text{top}(1), 1)$, $\text{share} = \text{most} \cdot \text{clamp}(1 - (\text{spin} + 0.6 \cdot \text{rise}) / \text{idleSpin}, 0, 1) \cdot \text{grip}$, up at most 0.3 a second (20b)

$\text{creep} = \text{share} \cdot (1 - p)$; 0 with the engine off or starting, in N, the brake over 0.02 or the handbrake on (20c)

Through the converter: the mod's A, or the game's automatic flag up (R too). By hand: a position of the mod's own gearbox, grip the clutch model's (0 pressed, 1 let up, 0 with its engine stalled), or a manual gear of the game's, grip 1. The converter's share reaches nothing at 0.87 of the gear's idle speed (idle/red · top). The governor's falls in a line from its most at a standstill to nothing at it: top(1) is the first gear's top without the low range, so the low range gets less (the game multiplies the throttle by the gear's torque), and rise is how fast the wheels gain, in rad/s a second over 0.2 s, while they gain. The look ahead and the paced rise keep it from outrunning the game's pull, which answers about a second late: given at once and whole, from 0.1 of idle to the red zone under idle, it overshot in the low range and cycled (the author's drive, 28 October). Nothing outside the world (a menu, the map, the garage, the pause), while the pull test runs, or for another truck. 0.15 is a first figure, for the test build's logs to set. On the models test's heavy truck (the game's pull at a standstill taken as 0.3 of its curve's) the creep holds 1.08 m/s in A and 1.22 m/s in first by hand, the gear's idle speed 1.49 m/s, the needle at idle both times. In L- with the game's pull half a second late it settles at 0.82 of the idle speed, and with the wheels free to spin at 1.00, without an overshoot (without the look ahead 2.1 times the idle speed, cycling). Gauges.log has a line a quarter second after each creep begins and one as it ends, four a second with Probe=1: the share and what it went by, the stub's calls for the truck and in all, the game's pull, the factor, its engine speed value (0.19 at idle) and its fuel rate.

Stuck in a gear picked by hand the engine stalls (`CreepStep`), at every `Gearbox` level, through the stall seconds' ramp of RequireClutch's stall:

stall: governing, grip ≥ 0.99; B4 rose $\leq 3\%$ over the last 0.5 s; spin < 0.25 idleSpin, gained < 0.03 idleSpin over 0.5 s; for 1 s (20d)

B4 is the game's pull (action +0xB4). The grace starts once it has built, so a slow truck pulling away is not stalled: a Zetros took 7 s to bring its wheels to a quarter of the idle speed, its pull rising all the while. A truck still gaining from rest is not stuck either. After the stall the creep stays off until the engine is off (3 s at the most), and Gauges.log says "creep: the engine stalled, stuck in gear N". Nothing through the converter, with the clutch held or on the brake.

Driving the game's gears

When the game's automatic flag (truck action +0x3C) is up in a forward gear, the mod takes over: it puts the flag down and from then on calls the game's own shift function, `ShiftGear(vehicle, gear)`, whenever its gear changes. The gear asked for is never under the lowest that fits. A gear the game picks (a tap, or A again) is put back, and so is a gear that appears within 2 s of such a pick. It hands the gears back on R, N or high, on a gear the player picks by hand (after 0.5 s), on a low range picked (the power factor 0.02 under its highest), with the setting off or the engine off.

29. Coolant and oil

A real engine's coolant takes about a quarter to a third of the fuel's heat, and the fuel goes with the pedal and the engine speed together. The model counts heat and cooling in one measure, 1 being a full load's heat (`models.h:397-647`):

$$\text{load} = p \cdot \text{speed}(\text{rpm})\ (\times 0.25 \text{ in neutral}),\ \text{heat} = 0.15 + 0.85 \cdot \text{clamp}(\text{load}, 0, 1) \tag{21}$$

$$\text{thermostat}(T) = 0.15 + 0.85 \cdot \text{clamp}((T - 82)/10, 0, 1),\ \text{fan}(T) = \text{clamp}((T - 98)/4, 0, 1) \tag{22}$$

$$\text{air} = 0.35 + 0.65 \cdot \text{clamp}(\text{km/h} \,/\, \text{AirFullKmh}, 0, 1) + 0.3 \cdot \text{fan}(T),\ \text{water} = \text{clamp}((\text{depth} - 0.5)/0.7, 0, 1) \tag{23}$$

$$\text{cooling} = (T - T_{out}) \,/\, (97 - 15) \cdot \text{thermostat}(T) \cdot (\text{air} + \text{water}) \tag{24}$$

$$T \leftarrow T + \text{rate} \cdot (\text{heat} - \text{cooling}) \cdot \Delta t, \text{ at most } 140\ ^\circ\text{C} \tag{25}$$

97 °C is where a fully loaded engine settles with all the air the radiator can get at 15 °C. depth is the mean water depth at the front wheels (wheel +0x17C): from about axle height (0.5 m) to the engine in it (1.2 m) the water does as much as full airflow. While the engine makes more heat than it loses, rate is r_{warm} up to 85 °C and r_{hot} from 90 °C, blended between; while it loses more, the coolant falls at 1 °C a second for each unit, whatever the settings, so a long overheat time does not make a hot engine slow to come back.

The three times in the settings set the rates (`TempRatesOf`, `models.h:511-525`). The way from one temperature to another at a rate of 1 is the sum, in quarter degrees, of 0.25/(heat − cooling); a rate is that way over the time it should take:

$$r_{warm} = \text{path}(15 \to 85\ ^\circ\text{C}, \text{load } 0.5, 13 \text{ km/h}) \,/\, \text{WarmUpMinutes},\ r_{hot} = \text{path}(90 \to 118\ ^\circ\text{C}, \text{load } 1, \text{standing}) \,/\, \text{OverheatMinutes} \tag{26}$$

$$r_{off} = \ln(\,(90 - 15)/(40 - 15)\,) \,/\, \text{CoolDownMinutes},\ \text{stopped: } T \leftarrow T_{out} + (T - T_{out})\, e^{-r_{off}(1\,+\,4\ \text{water})\,\Delta t} \tag{27}$$

Load 0.5 at 13 km/h is the mean of a drive logged in the game on 2026-10-08 (0.49 at 13.1 km/h). At the defaults (3, 1.5 and 15 minutes) r_{warm} = 0.735, r_{hot} = 0.966 and r_{off} = 0.00122 a second.

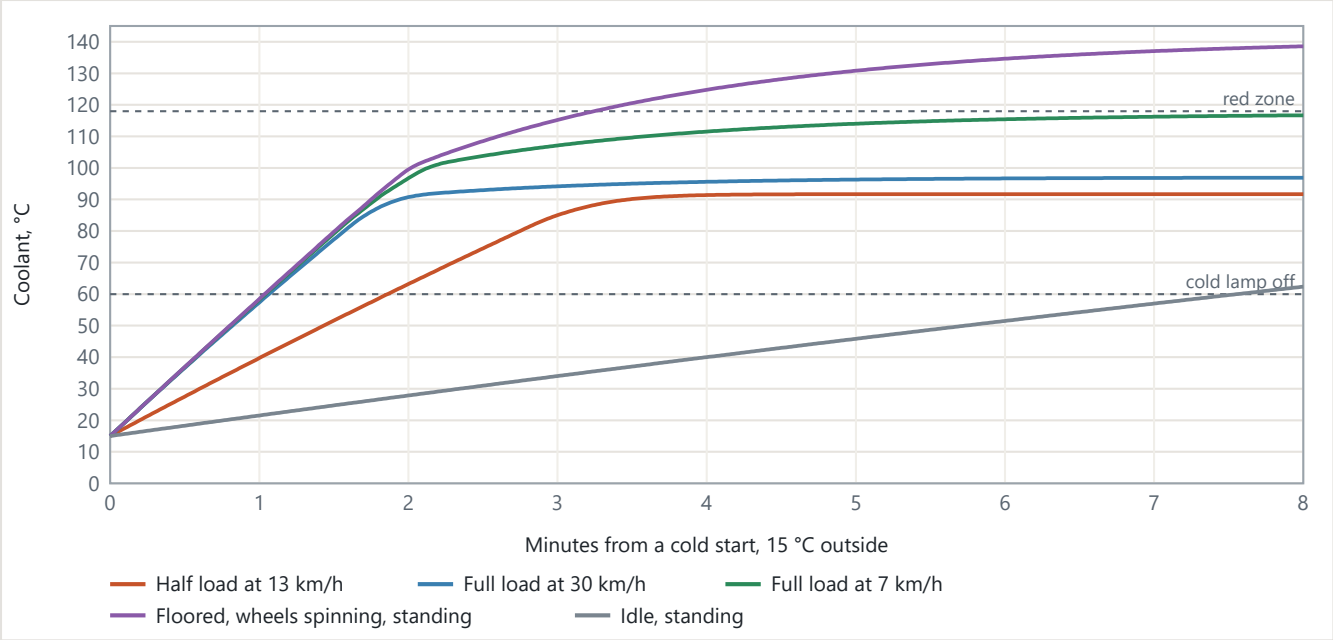


Figure 7. The coolant from a cold start, default settings.

Table 10. Where the coolant settles, default settings, 15 °C outside

Driving	Settles at
Idle, standing	86.1 °C
Half load at 13 km/h	91.7 °C
Full load at 30 km/h or faster	97.0 °C
Full load at 7 km/h	117.3 °C
Full load, standing	none: rises to the 140 °C cap

The oil heads for the coolant's temperature plus what friction and the pistons add, and follows with a lag:

$$T_{oil} \to T + 4 \cdot \text{speed(rpm)}^2 + 10 \cdot \text{clamp(load, 0, 1)} \tag{28}$$

The lag is found by bisection over a simulated warm-up, so that the oil reaches 85 °C at max(OilWarmUpMinutes, WarmUpMinutes + 1 min); at the defaults it is 102 s. A stopped engine's oil cools at half the coolant's rate. The cold lamp lights under 60 °C; the oil warning comes on at `OilWarnC` (120 °C) and goes off 2 °C under it, with a beep at once and every 4 s.

The mod remembers 16 kinds of truck. A kind driven again has cooled as a stopped engine for the time others were driven; a new kind starts at the outside temperature. Two trucks of one kind share a temperature.

30. Engine damage

Since 0.17.0 the gearbox takes damage too: a stall the mod's own gearbox orders for a shift without the clutch costs StallDamagePercent of what the gearbox takes (0 to 25, 5 by default, with Damage on), dealt on the game's thread as below under "Changed in 0.17.0". So does a burning clutch (ClutchBurn, with Damage on): its heat q rises by P = max(carried, 0) · Torque · slip over kClutchHeatCapacity (1.6e7; slip in rad/s), times the speed's share (1 under 5 km/h, 0.2 from 15 km/h, linear between: a slip while rolling is a short event, a truck pinned at a crawl is the abuse), and falls by kClutchCooling (0.173) of itself a second, half in 4 s, at any speed. A warning box (amber CLUTCH HOT) shows from kClutchWarnFrom (1.6); past kClutchBurnFrom (3.2, the box red, CLUTCH DAMAGE) and only while the clutch still slips under load it costs the gearbox q/3.2 · kClutchBurnRate (0.025) of its capacity a second; locked or open it costs nothing however hot, and what it owed is dropped. A repair (the gearbox's damage read falling) sets q to 0 and drops a burn's points not dealt yet. On the clutch model's own states (automatic, floored in third from a standstill on the test truck): amber after 1.6 s, red after 4.0 s; fourth red after 3.4 s, a gear the truck cannot pull after 2.9 s, first never amber (0.66 at most). On the idealised slip red after 1.80 s

and 0.073 in the two seconds after; three times the torque 0.50 s. From the end of a red slip the box is amber again after about 2 s and gone within about 10.

Two rates, in percent of the engine a minute (`models.h:2189-2247`):

$$\text{hot} = \text{HotDamagePercentPerMinute} \cdot (T - T_{\text{from}})/(125 - T_{\text{from}}) \text{ for } T > T_{\text{from}}, T_{\text{from}} = \text{HotDamageFrom (100 to 124 °C, 118 by default)} \tag{29}$$

$$\text{cold} = \text{ColdDamagePercentPerMinute} \cdot (60 - T)/60 \cdot ((\min(\text{load}, 1) - 0.5)/0.5)^2 \text{ for } T < 60 \text{ °C and load} > 0.5 \tag{30}$$

$$\text{points} = (\text{hot} + \text{cold})/100 \cdot \text{capacity} \cdot \Delta t/60 \tag{31}$$

At the defaults (10 and 5): 10% a minute at 125 °C, 31% at the 140 °C cap; a cold engine floored at the red zone at 0 °C loses 5% a minute, and nothing at moderate revs, as it burns little fuel there. Fractions are carried over and whole points written; nothing counts while the game window is in the background.

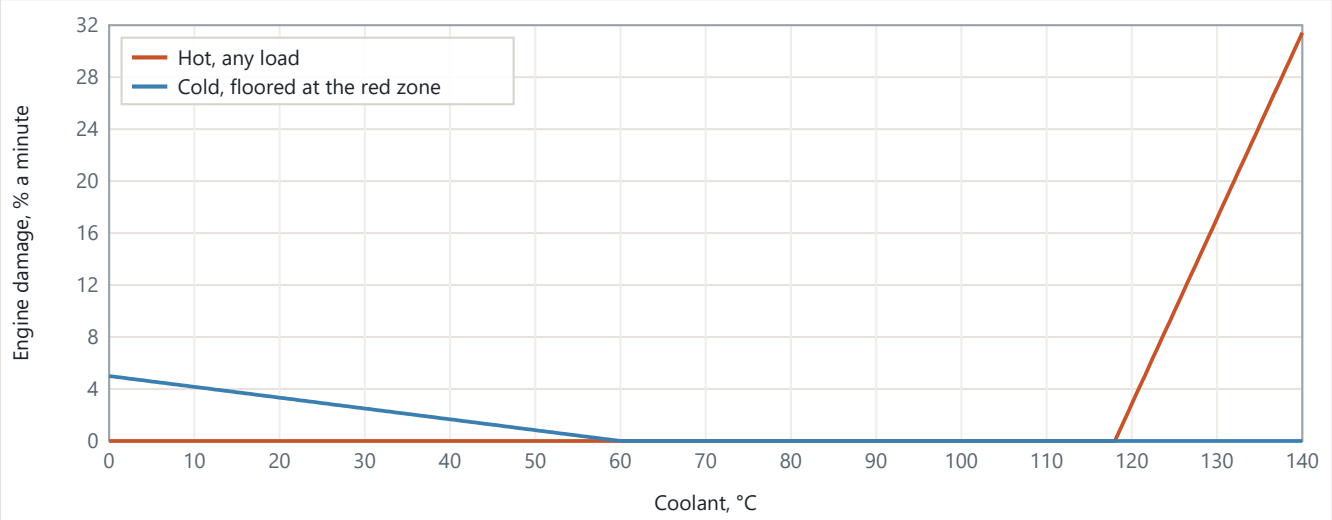


Figure 8. Engine damage by coolant temperature, default settings.

The damage goes into the game's own engine damage (the block behind vehicle +0x150; the handle with its second byte cleared, its flags at +0x10 with bit 0 set and bit 2 clear, data at +0x38: damage, capacity, effect). A write goes out at most once a second, only with the game in front, no menu open and after half a second of driving; control, vehicle and action must be unchanged, the capacity between 2 and a million and matching the engine's record. It stops one point short of the capacity: the last point stops the engine, and that stays the game's to do. With the engine's record found, it also writes the effect the game would, by the game's rule (Steam RVA 0xd6f690):

$$s = \text{damage}/\text{capacity}; \text{ if } s > \text{threshold}: e = (s - \text{threshold})/(1 - \text{threshold}), \text{torque factor} = (1 - e) \cdot T_{\min} + e \cdot T_{\max} \tag{32}$$

threshold, $T_{\min}$ and $T_{\max}$ are the engine record's (+0x74, +0x78, +0x7C). Without the record only the damage is written and the effect follows at the game's next damage event. The lamp is steady past half the capacity and blinks while damage is being added.

31. Boost and the turbo

The turbo is review 12's model as its addendum corrected it (`BoostStep`). It has two states: the rotor's speed *w*, as a share of what makes the most boost, and the manifold's pressure *b* in bar, gauge. *n* is the engine speed over the red zone's rpm, past 1 on the governor; *p* is the pedal and *q* the fuel the engine burns, as a share of full load at *n*.

$$\text{full}(n) = \text{most} \cdot \text{clamp}((n - n_0)/(n_1 - n_0), 0, 1) \tag{33}$$

$$q = \min(p, \text{limit}_{\text{last}} \cdot \text{cold}) \cdot \text{taper}(n); \text{ in N or with the clutch pressed } 0.25\ p \cdot \text{taper}(n) \tag{33a}$$

$$\text{diesel: } w_{want} = \sqrt{(\text{full} \cdot q^{1.6} / \text{most})};\ \text{petrol: } MAP = 0.3 + (1 + \text{full} - 0.3)\ p,\ w_{want} = \sqrt{(\max(MAP - 1, 0) / \text{most})} \tag{33b}$$

$$w \leftarrow w + (w_{want} - w)(1 - e^{-\Delta t/\tau}),\ \tau = \tau_{red} / \max(q\ n^4, 0.02) \text{ rising, } 1.5 \text{ s falling} \tag{33c}$$

$$b \to \min(\text{full}, \text{most}\ w^2) \text{ a diesel; } MAP - 1 \text{ while} \le 0, \text{ else } \min(MAP - 1, \text{most}\ w^2) \text{ a petrol; with } 0.1 \text{ s}$$

$$\text{limit} = \min(1, r\ (1 + b) / (1 + \text{full} \cdot \text{taper}^{1.6})),\ r = 1.2 \text{ a diesel, } 1.0 \text{ a petrol} \tag{34}$$ (33d)

taper(*n*) is 1 up to the curve's last point and past it the curve over that point's torque, the governor's cut. MAP is a petrol's manifold in bar absolute, set by its throttle: 0.7 bar of vacuum at idle, 0 at about half the pedal on 1 bar, the most floored. With the engine off or starting $q = 0$ and the manifold heads for 0. The limit is the powertrain's boost in equation 11. Settled and floored it is 1 at every speed, as the curve is a turbo engine's full-load curve, measured with the boost it makes there; only the spool costs (review 12, finding 1). On the governor full(*n*) is taken as the taper's fuel, as the curve already falls there: compared with full(*n*) alone, as the addendum writes it, the limit would cut there a second time. Two stages drop the boost a tenth for 0.3 s as *n* passes 0.6. The defaults per class are review 12's (the C13 and the 3406E for a truck), its τ_{red} fitted by the addendum's simulation to the review's bounds; `BoostMax` takes the place of the class's most.

Table 11. The turbo of each class

Class	Fuel	Most	n0 to n1	τ_red
Truck	diesel	1.8 bar	0.36 to 0.6	0.045 s
Scout	petrol	1 bar	0.2 to 0.3	0.01 s
Heavy	diesel	2 bar	0.37 to 0.68	0.085 s

Table 12. The turbo profiles (BoostProfile 0 to 3)

BoostProfile	n0 to n1 (a truck)	τ_red	Most ×
Standard	0.36 to 0.6	0.045 s	1
Small and quick	0.36 to 0.48	0.0225 s	1
Big and slow	0.504 to 0.84	0.0675 s	1
Two-stage	0.36 to 0.45	0.045 s	1.3

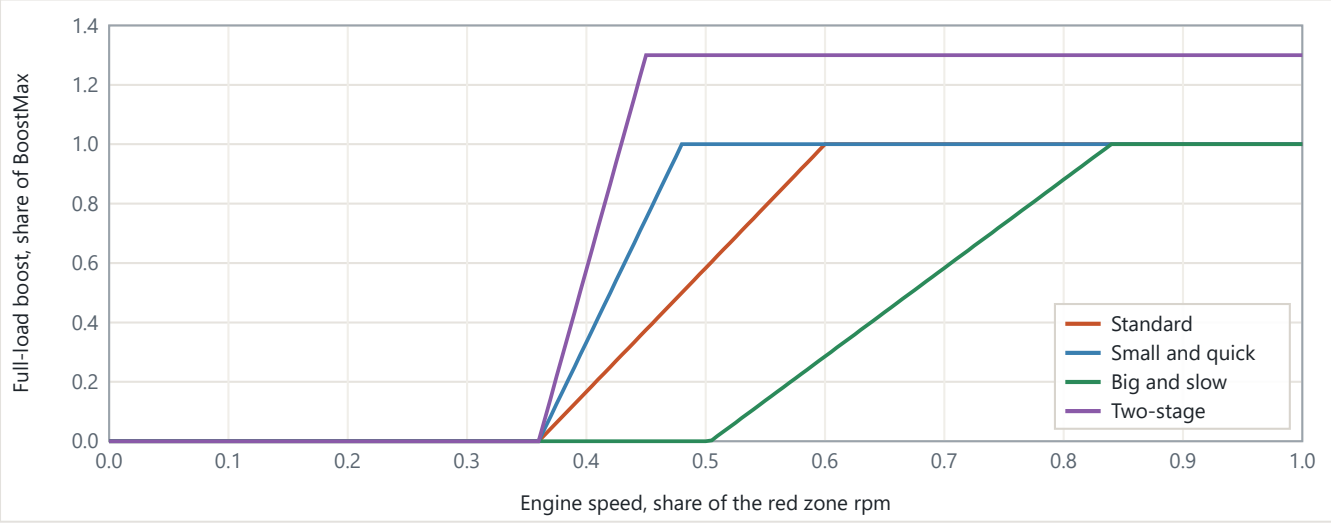


Figure 9. Where each turbo profile makes its boost floored, on a truck.

What the model gives, as the tests check it. Settled: a truck's diesel 0.00 bar at idle, 0.26 at a light foot (0.3) at 0.6 of the red zone's rpm, 0.67 floored at 0.45 and 1.80 at 0.8; a petrol scout −0.70, −0.19, 1.00 and 1.00 bar. At 0.6 of the pedal and 0.8 of the red zone's rpm the limit is 0.77 on a truck and 0.66 on a scout, over the pedal. From a light foot (0.2) floored, 90% of the torque comes in 2.0 s at 1100 rpm and 0.9 s at 1760 on a truck, 2.9 s at 1000 and 1.5 s at 1520 on a heavy truck, 1.4 s at 2000 and 0.3 s at 4400 on a scout. After a half-second lift at 0.8 a truck has 1.11 bar 0.1 s after the press (the limit 0.90) and all of it within a second; after three seconds 0.08 bar,

then 0.52, 1.29 and 1.80 at 0.5, 1 and 2 s. A petrol scout's manifold is at −0.69 bar on the lift and 0.22 bar 0.1 s after the press. Floored from idle through the converter the limit dips to 0.44 and is whole again at 1.5 s on a truck (0.41 and 2.4 s on a heavy truck). Floored in neutral a truck holds 0.20 bar.

The sounds come from the pedal over the last 0.25 s. A lift of half the pedal with at least 0.22 BoostMax built up is a blow-off; a lift of a quarter, the pedal then still, with at least 0.11 BoostMax, a flutter; then 0.8 s of quiet. Loudness is clamp(bar at the lift / BoostMax, 0, 1)$^{1.5}$; at 0.5 or more a blow-off plays the high recording and a flutter the middle one. Gain = volume × (0.2 + 0.8 loud), pitch ±6% at random. A diesel has no throttle plate and no blow-off valve: unless `DieselFlutter=1` its lift asks for the whistle (`WhistleSoundOf`), which holds only the player's own `turbo_whistle_` files, at the same gain and at a pitch of 0.5 + 0.5 · clamp(1.5 (w − 1/3), 0, 1) by the rotor at the lift; without them the lift is quiet (a whistle made by code was dropped after the author's drive of 8ED18842). The pedal going down past half fades a ringing sound out in 0.08 s.

32. Fuel range

The range is the fuel left over what the last few kilometres cost (`RangeStep`, `models.h:262-332`). Moving at 0.5 m/s or more, the way driven and the fuel used are marked every 50 m, up to 64 marks, the oldest dropped while the window still spans 3 km. A step that burns more than 0.1 l counts as filled or leaked, not burned. Once the window spans 500 m (a shorter way is mostly the pulling away):

$$\text{per km} = (\text{used} - \text{used}_0) \,/\, (\text{way} - \text{way}_0) \cdot 1000, \text{smoothed with } \tau = 10 \text{ s; range} = \min(\text{fuel} \,/\, \text{per km}, 999 \text{ km}) \tag{35}$$

Fuel burned standing is not charged to distance but lowers the fuel and so the range. Another truck kind or tank size starts it over; until 500 m it shows "---".

After the range the fuel use: the game's own rule (`TruckFuelRate`, litres a minute, 0 with the engine off), through a one-pole filter of τ = 1 s so it does not flicker with the pedal; one decimal, US gallons an hour (× 60 / 3.785) with Mph, "---" while the rule's numbers are not read. `ShowFuelUse=0` leaves it out.

33. Brake air

The engine drives a compressor that fills the tanks; a governor runs it from 6.9 bar up to 8.2 and lets it rest in between (`AirStep`, `models.h:2092-2160`):

$$\text{gives} = 0.23 \cdot \text{speed(rpm)} \cdot \max(1 - \text{bar}/12, 0) \cdot \Delta t \text{ while it runs} \tag{36}$$

$$\text{used} = 0.35 \cdot \max(b - b_{\text{last}}, 0) + 0.4 \text{ (handbrake let go)} + (0.1 + 0.25 \text{ while braking}) \,/\, 60 \cdot \Delta t \tag{37}$$

A full brake application takes 0.35 bar, letting the handbrake go 0.4 (the air that holds the spring brakes off), and the system leaks 0.1 bar a minute, 0.35 with the brake held. From empty the tank fills in 60 s at the red zone and about 3 minutes at idle; eight full applications with the engine off take a full tank to 5.4 bar, the red zone's edge (5.5).

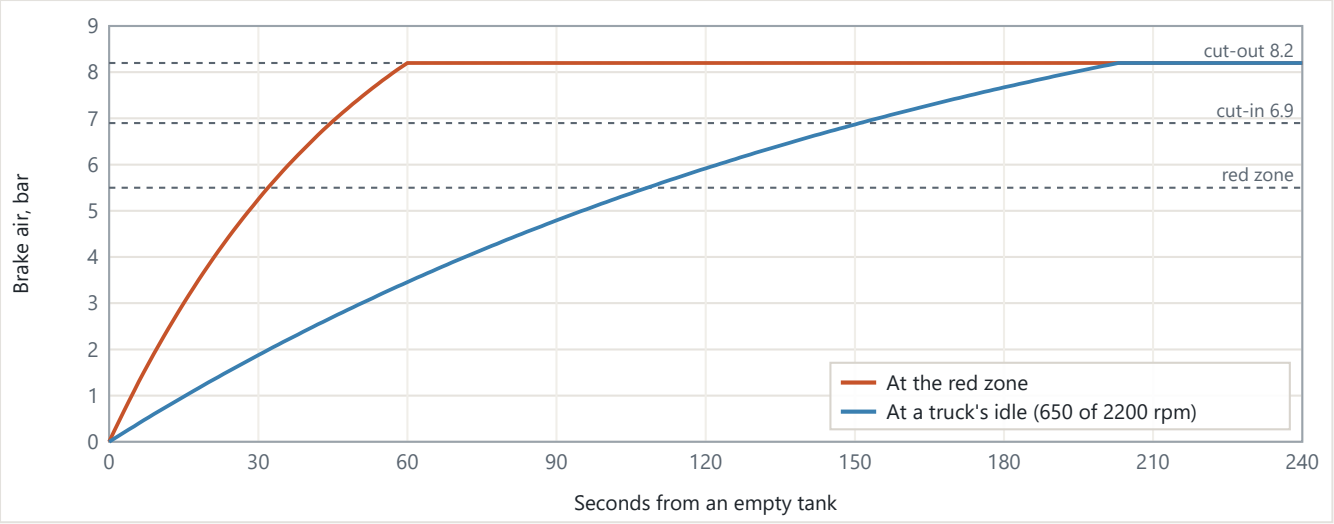


Figure 10. The compressor filling the tank, engine running.

34. The tilt lamp

The world-up parts of the chassis' forward, up and left axes (chassis +0x170, +0x180, +0x190) are smoothed with a 0.3 s time constant, so a bump does not light the lamp (`models.h:334-386`):

$$\text{pitch} = \operatorname{asin}(\text{forward}_y),\ \text{roll} = \operatorname{asin}(\text{left}_y)\ \text{(on its roof: } \pm 180° - \text{roll)} \tag{38}$$

Red from `RollDangerDeg` (25°) of roll or `PitchDangerDeg` (35°) of pitch, amber from `RollWarnDeg` (15°) of roll; pitch has no amber. Each stays until 2° under its angle, so it does not flicker.

35. Lamps and dials

Table 13. The lamps (gauge_draw.h:729-777)

Lamp	Reads	Lit	Colour
Handbrake	truck action +0x48	set	red
Headlights	vehicle +0x768 bit 2	set	green
All wheel drive	truck action +0x49	set	green
Differential lock	truck action +0x4A	set	amber
Low gear	gear 1, game automatic off, power factor	L−, L, L+	amber
Engine damage	the engine damage block	past half; blinks while adding and 2 s after	amber
Cold engine	the coolant model	under 60 °C	blue
Tilt	chassis axes	see above	amber, red
Oil	the oil model	blinks 400 ms at OilWarnC	red
Tire	the tire mod's block	see Shared memory	amber, green, red

Dials: fuel red under 10%; coolant 50 to 130 °C, blue under 60 and red from 118, with the oil as a second, amber needle; the oil dial 50 to 150 °C, red from `OilWarnC`; boost from 0 (a diesel) or −1 bar (a petrol engine) to 2 bar, or 3 for a turbo of more than 2 (× 14.5 for psi); brake air 0 to 10 bar, red under 5.5, hidden when the scales' class is scout (scouts brake by hydraulics); the clock, twelve hour marks and two hands over the game's hour (no hands while the clock is not found), with the outside temperature in force as a number in its face. The speed is the chassis velocity along its forward axis, × 3.6 for km/h. Every dial sweeps from 150° to 390°.

36. The outside temperature

With `AmbientByMap=1` the mod uses the first section that exists in `Gauges.ini`: `[Map:<map name>]`, then `[Map:<region>]`, then `[Map:<level id>]`; otherwise `AmbientC` of `[Gauges]`. A section gives `AmbientC`, a mean that

swings 10 °C (5 either way), or both ends, `DayC` and `NightC`. The day follows the game's clock (`DayNightC`, `maps.h:130-143`), with *h* the hour:

$$5 \le h \le 15\text{: share} = ½ - ½\cos(\pi(h-5)/10)\text{; otherwise: share} = ½ + ½\cos(\pi \cdot \text{since}/14),\ \text{since} = \text{hours after 15:00} \tag{39}$$

$$T = \text{night} + (\text{day} - \text{night}) \cdot \text{share} \tag{40}$$

Without a clock the mean holds.

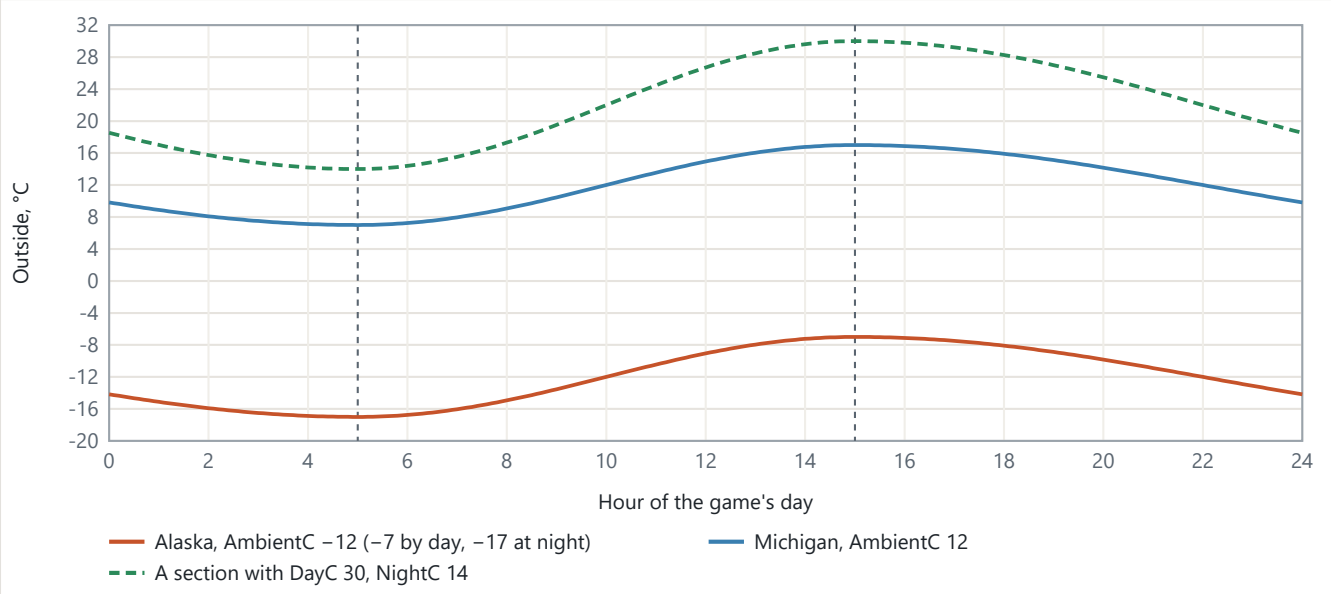


Figure 11. The daily curve: coldest at 05:00, warmest at 15:00.

Table 14. The regions' first values, written into an ini without map sections (maps.h:15-21): a guess for the player to set

Region	Mean	Region	Mean	Region	Mean
Michigan	12 °C	Maine	10 °C	Quebec	0 °C
Alaska	-12 °C	Tennessee	20 °C	Washington	10 °C
Taymyr	-20 °C	Ontario	5 °C	Chiapas	26 °C
Kola Peninsula	-10 °C	British Columbia	3 °C	Belozersk Glades	18 °C
Yukon	-15 °C	Scandinavia	-4 °C	Zurdania	22 °C
Wisconsin	8 °C	North Carolina	22 °C	Switzerland	-10 °C
Amur	2 °C	Almaty Region	24 °C		
Don	16 °C	Austria	8 °C		

Which map

The game does not say which map it is on in a place the mod reads, so the mod reads the newest save every 5 seconds for its `"lastLoadedLevel"`: Steam's `CompleteSave*.cfg` under `userdata\<account>\1465360\remote` (the Steam folder from the registry, or five folders up from the game), Epic's `CompleteSave*.dat` under `Documents\My Games\SnowRunner\base\storage`, and the Microsoft Store's files under `%LOCALAPPDATA%\Packages\*SnowRunner*\SystemAppData\wgs` whose first 95 bytes hold "CompleteSave". A save over 64 MB is skipped. Map names come from the game's `strings_english.str`.

The game's clock

In the Steam build the hour is a float at [exe + 0x2a8eb48] + 0x150, found in the running game on 2026-10-10. On another build a signature gives the static and the same +0x150 is assumed; a `[Clock]` section with the build's stamp overrides both. A reading counts in 0 to 24.5 and is taken modulo 24; unreadable for about 5 minutes of driving, the place is forgotten.

With `Probe=1` and no clock known, the mod looks for one while you drive (`clock_scan.h`): it follows the pointers in the exe's statics from 0x2a80000 to 0x2ac0000 two levels down (0x800 bytes of up to 4096 objects, then 0x200 bytes of up to 16384) and keeps every float in (0.0001, 24.5) and every double, up to a million slots (about 50 MB). It looks again every 10 s of the game's own time. A slot is steady after 4 looks at an even pace (within 10%) between $2.3 \cdot 10^{-5}$ and 0.08 units a second, a day of 5 minutes to 12 hours; values above 1.05 are hours, others a share of the day. A jump of an hour or more off pace, then two looks back on pace, marks the clock: skipping the night (T) or waiting in a garage makes that jump. The first jumped steady slot is the clock, or the only steady one after 180 s; it is saved to `[Clock]`.

37. Getting into the game

The loader loads the mod; the overlay draws in the game's Present, and everything else happens on the mod's own thread. The mod compares the exe's time stamp with the Steam build's. On that build it uses the places it knows; on any other it looks for them with signatures harvested from the Steam build (harvest 3, `game_globals.h:45-66`): each must match exactly once, and two that point to different places count as not found. Control, the wheel class and the game object are needed: a look that misses one is made again every 2 s, 31 looks in all (a mod loaded before the game's code is ready, as through another mod's early DirectX probe), then the mod stands down and the tab says so; the map, viewer, garage, input, engine sets and clock are optional. The objects' offsets are taken as they are on other builds.

Table 15. The places the mod reads from (truck.h:57-58, game_globals.h:24-29)

Place	Steam build RVA
Control (the truck you drive, +8)	0x2a8eb78
Wheel class	0x2258918
Game (menus, loading)	0x2a8eb90
Map screen	0x2a8eb30
Viewer	0x2a8eb38
Garage	0x2a90728
Input	0x2a8ebe0
Engine sets	0x2a9e8e0
Clock static	0x2a8eb48

Table 16. The fields it reads, and the two it writes (truck.h)

Object	Offset	What
vehicle	+0x768	flags: engine on, started, headlights, high beam, reversing, brake lights (bits 0 to 5)
vehicle	+0x114, +0x11C, +0x730	simulated seconds, stall seconds, start timer
vehicle	+0x150, +0x148, +0x158	engine, gearbox and tank damage handles
vehicle	+0xE38 to +0xE6C	fuel burn numbers
vehicle	+0x930, +0x968, +0x9C0 (+0x9A0)	engine set names, engine fitted, gearbox fitted (the truck's default gearbox)
vehicle	+0x60, +0x68, +0x200	model, truck action, wheel models
truck action	+0x38, +0x3C, +0x44	power factor, automatic flag (written), throttle (forward is negative)
truck action	+0x48, +0x49, +0x4A	handbrake, all wheel drive, differential lock
truck action	+0x58, +0x70, +0x74	gear list, gear engaged, gear
truck action	+0xB4, +0xCC	the game's pull, torque factor (written)
model	+0x5E8, +0x5F0, +0x588, +0x784	fuel, tank, kind, mass
chassis (model +0x230)	+0x170, +0x180, +0x190, +0x1A0, +0x230	forward, up, left axes, position, velocity
wheel (model +0x2C8)	+0xE8, +0x13C, +0x16C, +0x17C	drive type, centre, turning speed, water depth
engine record	+0x60, +0x68, +0x6C, +0x80	Torque, MaxDeltaAngVel, FuelConsumption, EngineResponsiveness
engine record	+0x74, +0x78, +0x7C, +0x84	damage threshold, torque factors, capacity

The engine's record is found the way the game finds it: through the engine sets list (entry: next +0, name +0x10, tree +0x30), a tree walk as the game's (RVA 0xc3c010), a text compare as its RVA 0x4d8230. Its fields were matched against a scout's file (Torque 195000). The mass was found by a scan of five trucks against their files.

Guards

- Every read goes through `ReadProcessMemory` and every write through `WriteProcessMemory` on the game's own process, and addresses below 0x10000 are refused: an object freed under the mod is a failed call, not a crash.
- The truck is found again every 500 ms and after any failed read; a read fails if control no longer points at the same vehicle. A truck must look sane: under 200 m/s, a tank under 100000 l with the fuel at most 1.01 of it, gears −1 to 16, throttle within ±2. A new kind is taken after 2 passes.
- The wheels are recognised by their class pointer; at most 32 wheels and 16 gears are read.

What it writes and calls

- With `Damage` on, the driven truck's engine damage, and with the record its effect and torque factor.
- With `Gearbox=1`, the automatic flag (truck action +0x3C), and it calls the game's `ShiftGear(vehicle, gear)`. That function is found by a pattern, also when another mod has hooked its start with a jump; it is called from the mod's thread inside an exception guard, and a fault turns it off for the session.
- With `Powertrain=1` and during the pull test, the torque factor (truck action +0xCC).
- With `Creep=1`, nothing in the truck: the game's wheels' drive is detoured (Creep at idle), at every `Gearbox` level, the first time a truck is driven with it on, and left in place until the game is started again.
- With all of them off it writes nothing.

Drawing

Next Gen's overlay (`ng_overlay.cpp`, the same file in Next Gen Tire Inflation) hooks Present, slot 8, and Present1, slot 22, on the game's own swap chains through method tables of their own. Where ReShade is installed,

ReShade's presents hand it each swap chain the game makes (`ng_reshade.cpp`); without ReShade it takes the game's imports of `CreateDXGIFactory1` and `D3D11CreateDevice` and gives each factory and swap chain they bring a method table of its own. DirectX's own tables are never written, so the Steam overlay, which hooks what those hold, never meets the overlay's Present. The release brings ReShade 6.8.0 without its start banner, with a `ReShade.ini` that loads no effects and keeps Home for ReShade's own window: on a tester's PC with RivaTuner, Discord's overlay and two monitors the game crashed without ReShade and not with it. Where ReShade is missing or does not take the overlay as an add-on, the start says so in the log, in a line at the top for 15 s that the first Next Gen mod in order draws, and in the menu above the tab's page (`NgReShadeCheck`). In the hook the mod draws with Dear ImGui 1.92.1, compiled in, on the game's device, through a render target view of the back buffer made for that frame; the game's render targets and state are put back before Present goes on. A 16-bit float back buffer (HDR, scRGB) gets the colours in linear light at a paper white of 203 nits.

The NEXT GEN menu is one window for every Next Gen mod in the game, kept in named shared memory (`Local\NextGen.Menu.<process id>`): the tabs, the one chosen, whether it is open, where it is, and its key (`NextGen.ini`, F2 by default). The mod whose tab is chosen draws it. Next Gen Graphics offers its switches through `NgGraphicsGetApi` (`ng_graphics.h`): the first Next Gen mod to claim it in the block draws a Graphics tab from them, and Next Gen Graphics silences its F keys meanwhile. While it is open the game's imports `GetCursorPos`, `SetCursorPos`, `GetAsyncKeyState` and `GetKeyState`, its window's mouse and key messages, and DirectInput's `GetDeviceState` and `GetDeviceData` for mice and keyboards give the game nothing new; releases still reach it.

The cluster is drawn into ImGui's background list, in the game's own font (TT Lakes Medium and DemiBold, baked with stb_truetype). One unit is the screen height / 1080 × `Scale`; it fades in and out over 0.15 s, and hides under 320 × 240 or when toggled off.

38. Shared memory

Named blocks of 64 bytes, one per game process, read without locks: the writer makes the sequence odd, writes, and makes it even again; a reader copies the block and keeps the copy only if the sequence was even and unchanged (up to 8 tries). A tick older than the limit means the writer is gone. Later versions only add fields at the end.

```
writer:  seq += 1 (odd)   barrier   fields   barrier   seq += 1 (even)
reader:  s = seq; if s odd, retry;   copy;   barrier;   if seq != s, retry
```

Table 17. Written: Local\SnowRunnerGauges.State.<process id> (state.h), every pass

Offset	Type	Field
0	u32	magic 0x47554147, "GAUG"
4	u32	version 1
8	u32	size 64
12	i32	sequence, odd while being written
16	u64	tick: GetTickCount64 at the write, ms
24	u64	vehicle (control +8), 0: none
32	f32	outside temperature in force, °C (always written)
36	f32	coolant, °C
40	f32	engine speed (the needle), rpm
44	f32	boost, bar
48	u32	flags: bit 0 vehicle, 1 temperature from a map section, 2 engine on
52	u32 × 3	reserved

The coolant, engine speed and boost are written only while you drive, otherwise 0. The tire mod reads the outside temperature from here.

Table 18. Read: Local\SnowRunnerTirePressure.State.<process id>, the tire mod's (tires.h); the fields used

Offset	Type	Field
16	u64	tick
24	u64	vehicle
32	i32	mode: 0 Low, 1 Reduced, 2 Normal, 3 Increased
56	u32	flags: bit 0 vehicle, 1 changing, 2 held, 3 too fast, 4 worn

Looked for every 5 s, stale after 2 s. The tire lamp shows only while the block's vehicle is the driven one: the letter L, R, N or +; red while the tires are too fast or one is worn out; otherwise blinking (400 ms) while the pressure changes or waits for the truck to stop, lit amber at Low and Reduced, green at Increased, dark at Normal.

Table 19. Read: a manual transmission mod's block, Local\SnowRunnerMT.State.<process id> (smt.h)

Offset	Type	Field
32	i32	gear (−1 R, 0 N, N+1 high)
36	i32	highest gear
40	i32	range (−1 low, 0 none, 1 high)
44	u32	flags: vehicle, A range, clutch held, clutch required, gear held, immersive, game shifting off, clutch on an axis, focused
48	f32	clutch 0 to 1
52	f32	power coefficient
56, 60	u32	stalls, shifts

Stale after 1 s. While that mod drives the truck in a manual gear, the mod's automatic stands down and a clutch model runs (engine speed from the clutch: free past 0.85 of the travel, full grip under 0.35, stalls under half idle); it is logged, not yet shown.

39. Files and what wins

- `Gauges.ini`, section `[Gauges]`: every setting. A missing ini is written with help text and the defaults; values that are not finite fall back to the defaults; tab edits are saved a second after the last one.
- `Gauges\trucks\<kind>.ini`, `[Truck]`: the seven scales (`TachClass IdleRpm RedlineRpm MaxRpm SpeedMax BoostMax BoostProfile`); a key present wins over `[Gauges]`. Written when one is changed on the tab while that truck is driven. The kind is the game's (model +0x588).
- `Gauges\gearboxes\<name>.ini`, `[Gearbox]`: `GearHold`, 1 to 3 s. The name is the gearbox fitted (vehicle +0x9C0), the truck's default (+0x9A0) only when none is read there.
- `Gauges\engines\<name>.ini`, `[Engine]`: `PeakTorqueRpm` (100 to 20000), `Aspiration`, `Fuel` (diesel or petrol), `TorqueCurve`.
- `[Map:...]` and `[Clock]` in `Gauges.ini`. Old `[Truck:...]`, `[Gearbox:...]` and `[Engine:...]` sections are moved into the files at load.

Scales: truck file over `[Gauges]`. The automatic's class: the truck file's if it sets a scale, else by mass. Gear hold: the gearbox file over the class (scout 1.5 s, others 2.5). Torque curve: `TorqueCurve` over `PeakTorqueRpm` over the class. The tab's sliders cover narrower ranges than the ini for some keys (Scale 0.5 to 2.5, warm-up 1 to 15 min, overheat 0.5 to 10).

40. Sounds

Five kinds: the turbo on a lift (high, middle, low), a diesel turbo's whistle on a lift (the player's own files only), the oil warning's beep, the air dryer's purge at each cut-out of the compressor (with the engine running, at 0.8 of the oil warning's gain), and the tab's buttons to hear the first two. For each the mod takes the player's own `sounds\turbo_high_1..4.wav` (and `_mid_`, `_low_`, `oil_beep_`, `air_purge_`, `turbo_whistle_`) next to the exe, else its built-in recordings (used with their author's permission; none for the beep or the purge), else sounds it makes; the whistle has neither, so without own files a diesel's lift is quiet:

- Blow-off: 0.7 s of noise through band-passes at 4.2 kHz (Q 0.7) and 1.9 kHz (Q 0.9), 6 ms attack, decaying with 0.13 s after 70 ms, high-passed at 700 Hz.
- Flutter: 0.6 s of puffs slowing from 26 to 13 a second, decaying with 0.22 s, band-passed at 1.3 kHz plus 0.35 × 3.4 kHz, high-passed at 400 Hz.
- Beep: two 0.13 s beeps at 2.2 kHz with a quarter of the third harmonic, 0.09 s apart, as a dashboard's buzzer gives them.
- Purge: 0.6 s of noise through a one-pole low pass at 4 kHz and a one-pole high pass at 600 Hz, a 3 ms attack, full for 0.1 s, then decaying with 0.15 s.

Every clip is brought to the same loudness (the loudest 0.1 s at RMS 0.1, peaks at most 0.6) and played at min(gain, 1) × 0.45 so two ringing together stay under full scale, never the same variant twice in a row, through XAudio2 2.9. No engine sound plays.

41. Limits and known issues

By design

- The game has no engine speed, temperatures, boost, brake air or range: all of them are models, tuned by drives logged in the game where the sections say so and otherwise the mod's own reasoning.
- The mod knows a truck by its kind: two trucks of one kind share a temperature, and 16 kinds are remembered. After a loading screen the truck that drives on under the game's id for it keeps its heat; with another truck, none before the load or none for a minute after it, every engine is cold again (see Changed in 0.17.0).
- The class by mass cannot tell every scout from a truck (the Fleetstar at 6314 kg is a truck; a modded scout of 9007 kg shifts as a truck until a truck file names it a scout). Mass is without addons, cargo or trailer.
- With `Gearbox=1` alone the pull is the game's; a gear's top speed stays the game's, and past it the game gives nothing.
- `Damage` has not been tried in co-op; leave it off there.
- On a build other than Steam's the clock is unknown until found with `Probe=1`, and the objects' offsets are taken as they are. A wrong clock place that still reads 0 to 24.5 is not noticed.
- A change of the game's engine damage that falls between the mod's read and write is lost.
- Probe logs run to 20 or 30 MB and the logs folder is never emptied. The pad's toggle buttons reach the game too.

Fixed after this manual's first version (0.15.8)

- Under the mod's automatic the needle followed the converter's engine speed without a cap and could run to the dial's end with the wheels spinning; now it goes no further past the red zone than wheelspin takes it elsewhere (1.1 of the way from idle to red).
- The brake air dial's red zone is drawn from the model's own figure (5.5 bar, as before).
- The README now says where another build stands down, how far a diesel comes down at a light foot, that the damage lamp also blinks after a write, and that the air dial is hidden on scout scales; the source comments say the engine and the automatic are handed to the game.

Fixed in 0.15.9, after a tester's keyboard drive

- Driven with keys, a lift shifted up at once and the next tap shifted back down: a key tapped to hold the speed swapped two close gears every second or two (19 of 23 lift upshifts taken straight back). Keys now wait 2 s and block lifts after one taken back, as above; wheel pedals and pads are unchanged.

Changed in 0.15.10

- Floored, the automatic shifts up where the next gear pulls more, or on the limiter: the cap at point(0.95) is gone. With the roof of 6 it sat just over where the next gear pulls more anyway (a ZIL-131: 3024 to 3056 rpm against 3072).
- An engine's own FullPull (the tab's Full pull up to): how far toward a gear's top it keeps its full pull, the roof following, so a petrol engine that makes its power up to its red zone can rev there (the ZIL-131 at 97: about 3150 rpm). Gone again in 0.17.0.

Changed in 0.16.0

- The mod is `Gauges.asi`, loaded by the ASI loader, and draws with Next Gen's overlay (Drawing, above); the release brings ReShade without its start banner, which hands the overlay the game's swap chains. The Gameplay tab is in the NEXT GEN menu, on F2, shared with Next Gen Tire Inflation. The mod is called Next Gen Gameplay; its files keep the name Gauges.

Changed in 0.15.11

- A look for the game's places that misses a needed one is made again every 2 s for a minute before the mod stands down. Loaded early through another mod's DirectX probe (a minimap mod), it looked before the game's code was ready and stood down.

Changed in 0.16.1

- RevCap (the tab's Rev cap 95%), off by default: with the mod's engine the pull ends at 95 percent of the way from idle to the red zone and the automatic shifts up there floored. For a truck that shoots from 40 to 80 km/h on its own, a game bug that RLM-like engines and this mod make worse. Gone again in 0.17.0.
- An engine's own Torque (the tab's Torque slider under Engine): how much of its torque the truck gets with the engine handed to the game, 10 to 100 percent (100 when not set). Lower it for an RLM-like engine with several times a stock engine's torque: at 25 an engine with four times the torque pulls like the stock one. The set-off gear follows it. The shift points stay: a constant share does not move where the next gear pulls more. The tab's torque plot is labelled Curve now.
- On another map every truck's engine is meant to start as cold as the air outside: the coolant and oil the mod remembers for each kind of truck are dropped when the game's save names another map. A reload of the same map keeps them. Known: this does not take effect yet. A drive on 25 October kept the engine warm over a map change. The code stays in and a fix follows.

Changed in 0.17.0

- A truck's gearbox file and gear hold go by the gearbox fitted (vehicle +0x9C0), not the truck's default gearbox (+0x9A0, which the mod read before): a truck with an upgraded gearbox was read as its stock one (a d110 on its highway gearbox, 26 October). A gearbox file named after the stock gearbox still applies while that one is fitted. The vehicle holds one block of 0x58 bytes a socket (the engine at +0x930, the gearbox at +0x988, the winch at +0x9E0): the list of the socket's types, the default part's name at +0x18 and the fitted one's at +0x38.
- The overlay copies a swap chain's method table that another program has put at the start of its memory. Before, it drew nothing there (no cluster, no panel and no NEXT GEN menu on two players' PCs). When a table still cannot be copied, `Gauges.log` says in whose code the table and its Present lie, so the program can be named.
- After a loading screen the truck that drives on under the id the game gave it keeps its engine's heat, and every other engine is as cold as the air outside: a travel between regions, a border crossing or a reload is no cold start, a game start, the garage, a recovery or a change of truck is. The cold start of 0.16.1 went by the

save, which names the new map seconds before a travel ends and minutes after a border crossing, so a travelled truck arrived cold and took cold-engine damage at once (a tester's UAZ, 26 October); the known issue is closed.

- The torque factor's roof is 6 for every engine again. An engine's FullPull and the tab's Full pull up to are gone: at 98 percent the roof was 23.8, and a d110 got a kick at the top of 2nd to 5th gear (25 October). RevCap and the tab's Rev cap 95% are gone too: the cap went by the engine's modelled speed, the kick by the wheels' share of the gear, and it never stopped one. For an engine with several times a stock engine's torque, lower its Torque.
- The mod finds its places on the Microsoft Store and Game Pass build the way Next Gen Tire Inflation does. Where its signatures miss, control is found by the truck update's three calls (way 2) or the same with the stack frame left open (way 3), the game's state beside control and the wheels' class by its type information (way 2 each). The log names the way. An optional place that is not found is left out with a line in the log that says what goes without it.
- SwapDials and the tab's Swap dials: the speedometer and the tachometer change places, in both layouts (asked for in the comments).
- HotDamageFrom and the tab's Damage from: where the hot engine damage starts, 100 to 124 °C, 118 as before; the rate is still reached at 125 °C (equation 29; from the mod.io list).
- The tab's Truck class keeps the scale numbers set by hand when the class changes; only the ones still at the old class's own follow the new class. The peak torque slider's tip says it ends at the red zone and the curve's peak sits at most at 0.95 of it.
- A clock on the cluster, ShowClock and the tab's Clock: an analog dial with the game's time of day, the outside temperature in its face, after the air dial in both layouts (from the mod.io list).
- The pad toggle (PadToggle) rests while the NEXT GEN menu is open, as the toggle key does; a toggle held as the menu opens or closes is no press.
- The air dryer's purge: a short hiss each time the governor stops the compressor at the cut-out, made by the mod's code, or the player's own sounds\air_purge_1.wav to _4.wav next to the mod; it follows SoundVolume.
- The fuel used a minute, after the range under the fuel dial, ShowFuelUse and the tab's Fuel use: the game's own burn smoothed over a second, litres a minute, gallons an hour with Mph (asked for in the comments).
- The game's gearbox functions are hooked through Microsoft Detours, calling through, as the ground for the mod's own gearbox (below); a probe line counts their calls. The manual transmission mod (SMT) is no longer needed and not compatible: the mod says so in the log and the tab when it finds it installed. The hooks go by the level: at Gearbox=0 none (the mod changes nothing in the game's code and runs as 0.16.x did); at 1 and 2 the eight gearbox functions; at 2 with Damage on also the game's truck update. Each is attached the first time a level needs it, also when the tab turns it on mid-game, through the same transaction and thread suspension, and is never detached while the game runs. On them the mod's own gearbox, in three levels of Gearbox: 0 the game's own; 1 the mod's automatic in A, the game's shifter the player's (a position picked there hands the truck back by the door it came through, AutoOffAndShift, ShiftToHigh, ShiftToNeutral or ShiftToReverse, whatever gear it names, so L+ is told from the automatic's own first gear and the automatic never shifts out of it; A takes it again) and only the game's reverse while the truck stands in A with the brake down and the game's automatic flag up refused (the flag is up only in the moment after the shift modifier is tapped or A picked again, before the mod puts it down; the hook reads it at the call), at ShiftToReverse or a gear below 0 through AutoOffAndShift or ShiftGear; with the flag down the shifter's R passes; 2 the mod's own gearbox, which takes the shifter: A, the numbered gears, N, R, H and the low range L-, L, L+ (0.2, 0.45, 1) as positions of its own, with a clutch (a button, or a pedal axis that the engine model runs behind; in these positions the needle and the torque curve handed to the game follow the clutch model's engine speed, TachFollow, as A follows the converter's; its pedal is a torque between the idle governor and the high-speed one, the curve's fall past the red zone, as a road truck's min-max governor; a stall of it with the clutch on an axis goes to the game's stall seconds as the gearbox's own does, a clutch on a button never stalls), a stall for a shift without it standing or crawling, under 1.5 m/s (RequireClutch: 5 written into the truck's stall seconds; at 1.5 m/s or more the gear grinds and stays out, N, with a log line; review 11), a held-gear mode for H shifters

with every gear bound (RequireGearHeld) and a choice to skip neutral (SkipNeutral, from L- to R), and down stopping at L- (DownStaysForward, SMT's immersive mode; it wins over SkipNeutral). Up and down walk an index: R -1, N 0, L- 1, L 2, L+ 3, gear g at g + 3, A at the game's gear + 3, H at the top + 4. At 2 the game's own ShiftGear, ShiftToAutoGear, ShiftToHigh, ShiftToReverse, ShiftToNeutral, AutoOffAndShift and SetPowerFactor are refused for the held truck (SetCurrentVehicle passes, and everything passes out of the world). Driven with a pad or the keyboard (Input 1 or 2) the first call of each door is read as a request for a position (R, N, H, A, gear g, or the low range once a SetPowerFactor under 0.725 follows a first gear; L+ reads as the first gear): the hook packs it into an atomic the mod's thread takes on its next pass, a request older than two passes is dropped, and the model applies it as that position's key under its own rules (RequireClutch stalls it too). With a wheel's pedals (Input 0) no request is made, and it runs as 1 until a binding that can be pressed reaches reverse (a key always, a wheel's while it is there, a pad's while a pad answers). While the NEXT GEN menu is open no action goes down and what was held stands; the reader steps every pass, and the pad and the wheels count only with the game in front. A or a gear picked from R while the truck still rolls back (speed under -0.5 m/s) waits in N until it stands, and the probe's letter is the model's while the mod's gearbox has the truck. The doors, as the first drives found them (2026-10-26, the d110 with Gearbox=0, every position by hand): H is ShiftToHigh, then AutoOffAndShift(7), then ShiftGear(7); R is ShiftToReverse, then AutoOffAndShift(-1), then ShiftGear(-1); N is ShiftToNeutral, then AutoOffAndShift(0), then ShiftGear(0); L is AutoOffAndShift(1), then ShiftGear(1), with no Shift-To function of its own; A is ShiftToAutoGear, then ShiftGear(1); each is followed by SetPowerFactor, 0.450 in L and 1.000 in A, H, N and R (L- and L+ wait for a truck that has them). With the game's automatic flag down the game makes no gearbox call at a standstill on the brake (8 s held, every count 0), so the brake-reverse is the game's automatic's and can only come while the flag is up; the shifter's R comes through the same door, which is why the flag and not the door tells them apart. Two bindings an action in Gauges.ini's [Gearbox] (Bind<Action>, Bind<Action>2: a key, a pad chord, a wheel's button or POV, or an axis for the clutch), bound in the tab by a press: the capture's first pass takes a snapshot of every control and it binds the first thing that changes from it, a control held or a POV pointing at the start left alone until let go and named in the log (closing the menu ends a capture; the menu's key, ToggleKey and the PadToggle buttons are never bound); defaults on the number pad, Num 8, Num 2, Num 5, Num 0 and Num 7 for up, down, N, R and A. SMT.ini's keyboard bindings (DirectInput scan codes, the number pad's 0x47 to 0x53 named as number pad keys) and options are taken over once, its DIFF LOCK and AWD too, and a default on a key it took is dropped. The manual transmission mod's state is no longer read.

- The differential lock and all-wheel drive on a key for every truck, the upgrade fitted or not, at every level of Gearbox (0 too): DiffLock and Awd in the bindings (no default), each press flipping one byte of the truck's action as SMT did, the lock at +0x4A and all-wheel drive at +0x49; the cluster's lamps read the same bytes. A key held is one press; nothing is written while no truck is driven or the NEXT GEN menu is open. A byte the game puts back to 0 on the next pass after a write of 1 is said once a truck in the log and left (`LocksStep`, `truck.h`).
- A shift without the clutch costs the gearbox: on each stall the model orders (RequireClutch), round(capacity · StallDamagePercent/100) points go into the gearbox's damage block (behind vehicle +0x148, laid out as the engine's), at most up to its capacity; 0 to 25, 5 by default, nothing with Damage off; a stall of the game's own costs nothing. As the tire mod deals tire damage, it is dealt on the game's own thread: the mod's thread posts a job (`StallJob`, `truck.h`); a detour on the game's truck update (Steam RVA 0xa56970, the update of 20 October 0xa58220; found by its call of GetForegroundWindow, then its first bytes before it; first argument an object with the vehicle at +0x20) deals it on its next frame for that truck, then runs the game's damage update (RVA 0xd6f690 and 0xd710b0, found by its first bytes), which sums the parts at vehicle +0x108 and brings up the truck card. The log gives the points and the sum the game reports after; a job no frame takes within 5 s is taken back. The probe's damage line names the gearbox's damage beside the engine's.
- A slipping clutch under load heats up and costs the gearbox, faster the stronger the engine (ClutchBurn, default 1, with Damage; the tab's "Clutch burns"): `ClutchHeat` in `models.h`, stepped from the clutch model each pass with the engine's Torque figure times the profile's share, the heat kept per kind of truck as the temperatures are; the burn's whole points go through the stall damage's job. The log says "clutch: burning, the gearbox pays" and, when it cools under the burn point, the points it cost; the probe's clutch line gives the

heat and its share of the burn point. The figures are in the damage section. Every clutch heats, a pedal's or the one that works by itself (on a button, or none). The automatic clutch rides as a driver's foot would: while it slips the engine goes to kAutoTakeUp (0.9) of the pedal's governor speed at the free engine's pace and the clutch passes on what the engine gives there, from none at idle to all of it a tenth of the way to the red zone, until the gearbox's input has come up to the engine; then it locks, and it lets go again when the input falls under idle. From kAutoLockKmh (15 km/h) up it stays locked and never slips; with the engine held under its idle speed in gear (lugging) the mod raises the game's stall seconds (vehicle +0x11C) by the pass's time and puts them back to 0 when the engine recovers, so the game's own "engine stalling" notice shows and it kills the engine at 1.4 by its rule (a drive confirms the notice). A stall under RequireClutch ramps the seconds from 0 to 1.5 over 0.3 s (StallStep) instead of writing 5 at once. The controls are read by a poll thread about 200 times a second and latched (InputLatch), so a tap between two passes still gives its press. Floored from a standstill (a truck of gears 3, 6, 12 and 16, the input gaining 2000 rpm a second a share in first and by the square of the gears' speeds in the others) first reaches 0.57, the box never shows, third goes amber after 1.6 s and red after 4.0 s, fourth red after 3.4 s and a gear the truck cannot pull after 2.9 s; it heats fully only under 5 km/h and a fifth as fast from 15 km/h, and cools to half in 4 s. Those figures are at ClutchHeatPercent 100; the tab's "Clutch heat rate" (ClutchHeatPercent, 10 to 300, 60 by default) scales the heating only: at 60 first reaches 0.40, third goes amber after 2.9 s and peaks at 2.16 (no red), fourth amber after 2.6 s (2.72, no red), and a gear the truck cannot pull amber after 2.4 s and red after 6.0 s. The cost only while it slips past the burn point; a repair clears the heat ("clutch: repaired, the heat is gone"). A stall's job waits while the same truck is driven and goes only when another truck is.

- Warning boxes in the shape of the tire mod's TIRE DAMAGE box (800 by 148 of its reference pixels, a stripe down the left, the title, a line of advice, the number at the right and a bar along the bottom), stacked below it at stage 52, 205 plus a slot of 148 + 16 (`WarnOrigin` in `gauge_draw.h`), drawn by the mod's own overlay with its own atlas of the game's font baked for the stage's scale, fading in and out over a quarter second, on the background list and never while the NEXT GEN menu is open. The clutch's (ClutchBurn and Damage on, slot 1): amber CLUTCH HOT "Ease off or shift down" from kClutchWarnFrom, red CLUTCH DAMAGE "Shift down, the clutch is wearing" from kClutchBurnFrom, the number q in percent of the burn point. The engine's (Damage on, slot 2): amber ENGINE HOT "Ease off, the coolant is in the red" from kTempRed (118 °C), red ENGINE DAMAGE "Ease off, the engine is wearing" while DamageHot is above 0, the number the coolant's degrees over 118 in percent of the way to 125. With HotDamageFrom at its default (118) the red follows the amber at once. At the turbo's full boost with the pedal floored the red title reads DANGER TO MANIFOLD with "Keep an eye on the floor pan", and once a point of the heat's damage lands under it, "The floor pan fell off" for the rest of that red spell.
- Known: the own gearbox is not tested on a wheel or shifter yet; a test on one follows. In a manual position the engine pulls by the mod's torque curve at the revs it is at, so a high gear at low revs pulls less than A keeps it pulling. A downshift at speed does not yet send the revs over the red zone: no engine braking from it and no over-rev damage. Lugging the engine in too high a gear at speed raises the game's own stall seconds; whether the game's stalling notice shows for it is unconfirmed. The low range's L- and L+ power factors are taken as 0.2 and 1.0 until a truck with them is driven at Gearbox=0; L is 0.45 as driven. On Windows 10 the NEXT GEN menu may not draw while ReShade loads the .asi files as add-ons: turn that off in ReShade's Add-ons tab (one player's report, under investigation).

Changed in 0.18.0

- The turbo of review 12 with its addendum: a rotor that spools by the fuel and the engine speed ($\tau = \tau_{red}/\max(q\ n^4, 0.02)$, down with 1.5 s), the boost by the fuel along each class's own full-load line, a limit that is 1 settled and floored (the air's share of the torque, kBoostTorque, is gone), a diesel or a petrol engine per class and per engine (`Fuel`). The author's drive and review 12, 28 October: the boost's share cut the torque at every speed below full boost even settled, and the dial followed the engine speed.
- The boost dial from vacuum on a petrol engine (−1 bar) to 2 or 3 bar by `BoostMax`, which now goes to 3; a diesel's lift is quiet as its turbo coasts down (own `turbo_whistle_` files play there; one made by code was tried and dropped), the flutter on a diesel a setting (`DieselFlutter`, off by default).

- Creep at idle (`Creep=1`, on by default; asked for on mod.io, 27 October): an idling truck in gear moves, through the converter to about walking pace, in a gear picked by hand at the gear's idle speed, through a detour of the game's wheels' drive that raises its throttle for the driven truck only. The pedal, the game's engine speed, fuel and sound stay at idle. At zero pedal the needle maps the wheels in proportion, never under idle, and the converter's stall rise needs the foot on. The governor falls in a line to the gear's idle speed, gives less in the low range, rises at most 0.3 a second and looks 0.6 s ahead while the wheels gain: at once and whole it overshot in the low gears, and on soft ground the wheels and the needle jerked (the author's drive, 28 October). Stuck in a gear picked by hand, at any Gearbox level, the engine stalls a second after the game's pull has built; in A it never does.
- The probe lines (review 11): "powertrain" ends with the factor's own terms (LiftShare's x, LiftRoof there, the pace) and marks its power and fuel as the model's own; "automatic" ends with what its rules decide by (feel, the points up and down in force, the lock and unlock points, the pause, wanted, the hold, the kickdown's arm, the cruise, lift and stuck timers, the start gear) and x against the model's own gear. The own positions' shift lines carry the clutch model's figures instead of zeros, and the hunt line is written only while the model drives. The fields the log tools read are kept where they were.
- Each gear's top is learned where the game's own pull reaches nothing (review 11): a floored plateau is logged and from then on gives that gear's top, kept in the gearbox's file, so the limiter, the floored points, the needle and the lift go by the gear's real top. The high gear's a + 2 holds.
- At Gearbox=2 in the gearbox's own positions there is one engine (review 11): the needle and the torque curve the game gets go by the clutch model's engine speed, not by the wheels' share and a pedal lift that worked with the clutch pressed too. The clutch model's governor is a road truck's min-max one, the pedal a torque between the idle and the high-speed governors. A stall with the clutch on a pedal's axis is handed to the game; a clutch on a button never stalls the engine as the truck moves off, as before (stuck at idle with the creep it does).
- RequireClutch: a shift without the clutch at 1.5 m/s (about 5 km/h) or more grinds and stays out, N with a log line, instead of stalling the engine; standing or crawling it still stalls (review 11: it stalled at 60 km/h too).
- With Powertrain=1 the lift's fade starts at 0.7 of a gear's top (from 0.8), so the push comes down over the last three tenths, less steeply. Within two seconds of a cut a push that grows comes back at about 1 a second; an ordinary rise through a gear, a push that falls, a new gear and the first write after the hand takes the truck come at once. A tester's log of 28 October: a heavy truck held near its top gear's limit at part throttle lost and regained its push every two seconds, the factor swinging 1.0 to 2.4 and the speed 36 to 39 km/h, since the fade was steep and the push came back in one jump. The top speed stays the game's own; the floored shift up moves with the fade, to about 0.905 of a gear's top with a truck's roof (from about 0.92).
- The automatic's torque converter is a truck's (review 11). It stalls at 0.8 of the red zone (at most 2600 rpm; 0.75 and 2400 in 0.17.1), the engine climbing from there through a launch, and couples at 0.87 with 4 percent slip. A launch locks once the converter has all but coupled, a shift made locked locks again on the next pass, and it locks in first gear too. Locked, it stays locked on a lift, so the engine follows the wheels and brakes the truck, and in engine mode a press never opens it: it lets go only under 1.4 idle with the foot on and 1.04 on a coast. The truck felt like a CVT or a rubber band: the converter opened on every lift and every shift, never locked in first, and the engine flared against its stall speed at each press (two players' reports and the author's own drives, 28 October; review 11 found a press from a low cruise still flared it).
- In engine mode the automatic cruises where a diesel pulls cleanly (review 11): at a light foot the converter locks from 1.65 idle (from 1.3) and the gearbox shifts down at 1.6 idle (from 1.15). A scout without a profile cruises from about 2370 rpm instead of 3300 (the light-foot shift-up point from the curve's 0.9 point, at most 0.45 red). On the limiter a gear going up is held 0.9 s, not the hold's 2.5 s. When the floor lifts the game's gear past the model's held one, the model's gear follows. The coast shift down with the brake on lands low for an engine brake; it matters once the mod has one.
- A binding's capture in the tab binds the first thing that changes. A control already held or a hat already pointing when the slot is clicked is left alone until it is let go, and Gauges.log names it. Two players could bind

nothing in the tab, and only Escape worked: the capture waited for every control to be at rest, and a controller reporting a hat or a switch as pressed kept it waiting.
- With Powertrain=1 the automatic's shift up out of first (and out of the gear it sets off in) goes by the wheels as they turn, spinning or not. A tester's log of 27 October: a ZIL-157 on dirt sat in first for 10 to 40 s with its wheels spinning near the gear's top, since the shift up went by the ground speed, and the game's pull falls by the wheel speed. From the second gear up a shift up goes by the ground as before. A truck stuck where it stands, under 0.3 of the start gear's top over the ground, keeps the ground's rule, so it does not shift up and back.

Changed in 0.17.1

- At Gearbox=1 a position picked on the game's shifter hands the truck back by the door it came through, whatever gear it names, so the automatic never shifts out of a player's L+ and the game keeps the diff lock. Before, the hand-back went by the gear number. L+ and the automatic's first are the same gear, so the automatic shifted on out of L+ and the game dropped the diff lock (a tester's report, 28 October). L and L- picked from the automatic's first gear were missed the same way.
- The engine's lift fades over each gear's last fifth: its roof is flat up to 0.8 of the gear's top and falls in a straight line to the game's own at the top. The roof is 6 for a truck or a heavy truck and 3 for a scout. A vanilla Defender 90 floored on a road shot from 27 to 58 km/h in 5 seconds through three gears (27 October). The floored upshift now comes earlier: a ZIL-131's at 2880 to 2928 rpm (was 3024 to 3056).
- A pad's stick directions bind as buttons: LSUp, LSDown, LSLeft, LSRight, RSUp, RSDown, RSLeft and RSRight, bits 16 to 23 of the pad's mask above XInput's 16 buttons, each down while its axis is past half its travel from the dead zone (8689, the right stick's, for both: 20728 of 32767). The capture and the poll thread's latch take them as buttons, so `Pad LB+RSUp` binds and a flick between two passes counts (asked for in the comments). A pad binding with an unknown button in Gauges.ini ended the game at its start (a 99-character reason copied with strcpy_s into 96 bytes); the reason is now cut to its room.
- At Gearbox=2 up and down walk the low range too: R, N, L-, L, L+, 1, 2 and on to the top, as SMT's did, on every truck (L+ is the first gear at 1.0, so it and the 1st are the same gear). SkipNeutral now steps from L- to R and back.
- DownStaysForward, off by default: at Gearbox=2 Gear down stops at L-; from L-, N or R it does nothing, and up from R still walks N and L-. It wins over SkipNeutral, and the once-only import of SMT.ini takes SMT's IMMERSIVE MODE over as it (asked for in the comments).
- At Gearbox=2 driven with a pad or the keyboard (Input 1 or 2) the game's shifter picks the position: the calls stay refused, and the first door of each shift is a request the model takes as that position's key (L+ reads as the first gear). With a key bound the slider used to move while nothing shifted. With a wheel's pedals (Input 0) the game's shifts stay refused and level 2 still runs as 1 until a binding reaches reverse (asked for in the comments and seen in the author's own drive).

Not changed

- The oil's lag is fitted with the engine speed set equal to the load (0.5).
- The light-foot downshift floor is 1.15 × idle: 748, 920 or 690 rpm by class.

42. Constants at a glance

Table 20. The models' constants

Constant	Value	Unit	Where it comes from
Pass	50 ms sleep; Δt 0.001 to 0.2 s		the mod
Class scales	truck 650/2200/3000 rpm, 100 km/h, 1.8 bar; scout 800/5500/7000, 120, 1.0; heavy 600/1900/2500, 80, 2.0		not stated
Gear top	2a + 5 rad/s, or learned on a floored plateau (pedal 0.9, B4 0.02, wheels 0.85 to 1.15 of it, 1% for 1 s); high a + 2		the game's, as read; review 11
Game pull	rise 2.8x − 0.25, fall 2.1(1 − x)		action +0xB4, floored, two trucks
Shift gain	5% more pull		not stated
Needle	floored standing 0.65, neutral 0.9, overrun 1.1; lag 0.18/0.25/0.4 s; dip 0.85 for 0.3 s		not stated
Stall, start	dies at 1.4 s (0.714/s); start 1.8 s		the game's timers
Pedal shape	p^1.8; up 0.6 s, down 0.15 s		"half way down gives 0.29"
Cold oil torque	15% at 0 °C, none from 60 °C		not stated
Boost limit	min(1, r (1 + b)/(1 + full · taper^1.6)), r 1.2 a diesel, 1 a petrol		review 12, finding 1, and its addendum
Torque factor	0.05 to 6 × the game's (a scout 3), the roof fading from 0.7 of a gear's top to the game's own at the top, a rise at most 1 a second within a gear within 2 s of a cut, times the engine's Torque share where set; step 0.005		a tester's log: the roof of 3 in 1133 readings; the roof of 6: a d110's kick at 23.8, 2026-10-25
Converter	stall 0.8 red, at most 2600 rpm, open + 0.15 input with the foot on; slip 1.04; coupling 0.87; ratio 2 at stall; lock 0.5 red (engine mode 1.65 idle light to 0.5 red floored), in the set-off gears at a speed ratio of 0.92; open 0.4 red (engine mode 1.4 idle, 1.04 on a coast); locked on a lift		Allison 3000/4000 data sheets (review 11); two players' reports, 28 October
Creep	converter 0.15 at a standstill to nothing at a speed ratio of 0.87; governor 0.15 × top/top(1) in a line to nothing at the idle speed, up 0.3/s, look ahead 0.6 s; brake 0.02; stall a quarter of the idle speed for 1 s once the pull rose ≤ 3% in 0.5 s		a first figure (review 11's addendum; the addendum to the creep and stall brief)
Shift timing	pause 1 s; up 0.4 s (0.15 floored); down 0.5 s; coast 0.3 s		"a short first gear is through in a second"
Cruise, lift, coast	cruise: pedal 0.5 or less, steady within 2%/s for 2 s; lift: gaining 2%/s at most, slowing 4%/s at most; coast point(0.15), braking point(0.25); with keys a lift waits 2 s, and one taken back within 6 s blocks lifts 10 s, doubling up to 60 s		a car's nine-speed automatic filmed on the road, 2026-10-19; the keyboard drive of a tester, 2026-10-19
Kickdown	part throttle 0.2 to 0.9 (pad 0.75) for 0.3 s, then 0.9		"a pad's trigger held near its end dips"
Gear hold	2.5 s, scout 1.5 s; on the limiter 0.9 s		Fleetstar 1 to 5 in 2 s; driven in the game; review 11
Light foot	up no lower than 0.8 × the curve's 0.9 point, at most 0.45 red; down not under 1.6 idle light, 1.15 floored		truck automatics' part-throttle points (review 11)
Hunt guard	6 s window, 8 s kept, up to 30 s		a tester's Tayga, 0.15.0
Class by mass	scout ≤ 6000 kg, heavy ≥ 16000 kg		TUZ 108, Tayga, Western Star
Coolant	cruise 97 °C; thermostat 82 to 92; fan 98 to 102; idle heat 0.15; still air 0.35; neutral 0.25		cruise by design; rest not stated

Constant	Value	Unit	Where it comes from
Warm-up path	load 0.5 at 13 km/h		a drive logged in the game, 2026-10-08
Defaults	warm-up 3 min, overheat 1.5, cool-down 15, air full 30 km/h, oil warm-up 5, OilWarnC 120		not stated
Water	from 0.5 m to 1.2 m; stopped × 4		"about axle height", "engine in it"
Oil over coolant	4 °C × speed², 10 °C × load; at least 60 s after the coolant		not stated
Engine damage	hot 10%/min at 125 °C from HotDamageFrom (118 by default, 100 to 124); cold 5%/min at 0 °C above load 0.5		not stated
Boost	truck n 0.36 to 0.60, τ_red 0.045 s; scout petrol 0.20 to 0.30, 0.010 s; heavy 0.37 to 0.68, 0.085 s; fuel^1.6; rotor down 1.5 s; manifold 0.1 s; petrol throttle shut 0.3 bar abs		review 12 (the C13 and the 3406E); τ_red fitted by its addendum's simulation
Turbo sounds	window 0.25 s; blow-off lift 0.5 at 0.22 BoostMax; flutter 0.25 at 0.11; quiet 0.8 s; a diesel's lift quiet, own whistle files pitched 0.5 to 1 by the rotor		not stated; a diesel has no blow-off valve (review 12, finding 6)
Range	marks 50 m, window 3 km, first 500 m, gulp 0.1 l, smooth 10 s, at most 999 km		"a shorter way is mostly the pulling away"
Brake air	cut-out 8.2, cut-in 6.9, flow 0.23 bar/s, most 12, brake 0.35, spring 0.4, leak 0.1 and 0.25 a minute		fill in 1 min at red, 3 at idle; 8 applications to the red zone
Tilt	smooth 0.3 s; back 2°; warn 15°, roll 25°, pitch 35°		"so a bump does not light the lamp"
Day	swing 10 °C; coldest 05:00, warmest 15:00		"coldest just before sunrise"
Clock search	10 s of game time a look; 4 steady looks; pace 2.3e-5 to 0.08/s; jump 1 h; 180 s alone		"a day of 5 minutes to 12 hours"
Shared blocks	64 bytes; tire stale after 2 s, gearbox mod after 1 s; looked for every 5 s		the mods' layouts
Sounds	RMS 0.1, peaks 0.6, voice × 0.45, pitch 0.25 to 2		"two that ring together stay below 1"

43. Licence and credits

GNU General Public License v3.0 (GPL-3.0-only). The cluster and the menu are drawn with Dear ImGui by Omar Cornut, and their text is baked with stb_truetype by Sean Barrett; ReShade by Patrick Mours (crosire), which the release brings along without its start banner (BSD 3-Clause licence), hands the mod the game's swap chains through its add-on API. The turbo recordings are used with their author's permission. SnowRunner is a game by Saber Interactive; this project is not affiliated with Saber Interactive or Focus Entertainment and contains no game files.

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